

Ethical Hacking, Fourth Edition - Lab 03

Amatun Haque

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15 hours, 0 minutes

100%

Report Generated: Friday, January 30, 2026 at 1:08 PM

Part 1: Identify Vulnerable Windows Systems

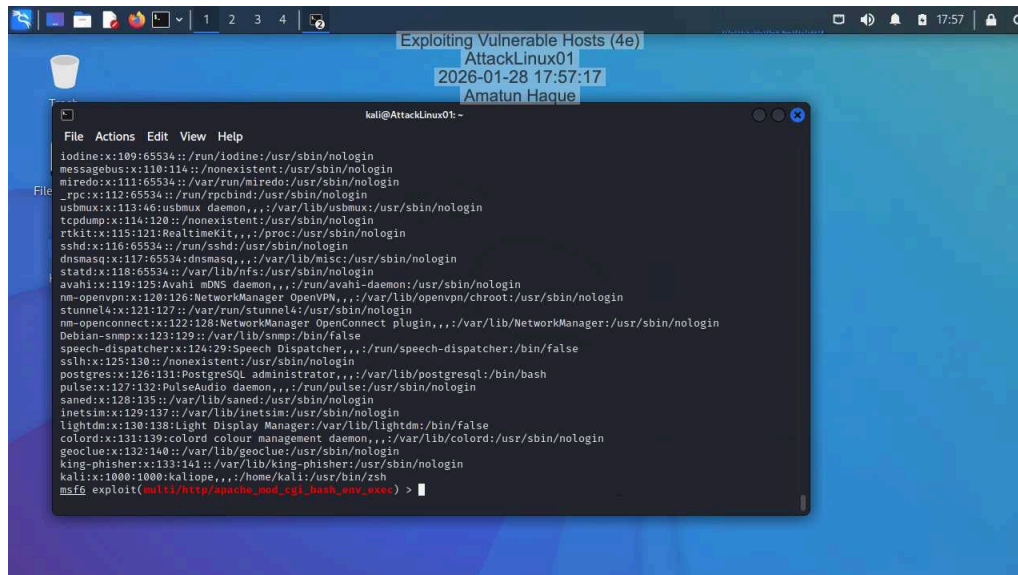
Operating System and Version

172.30.0.40Microsoft WindowsWindows Server

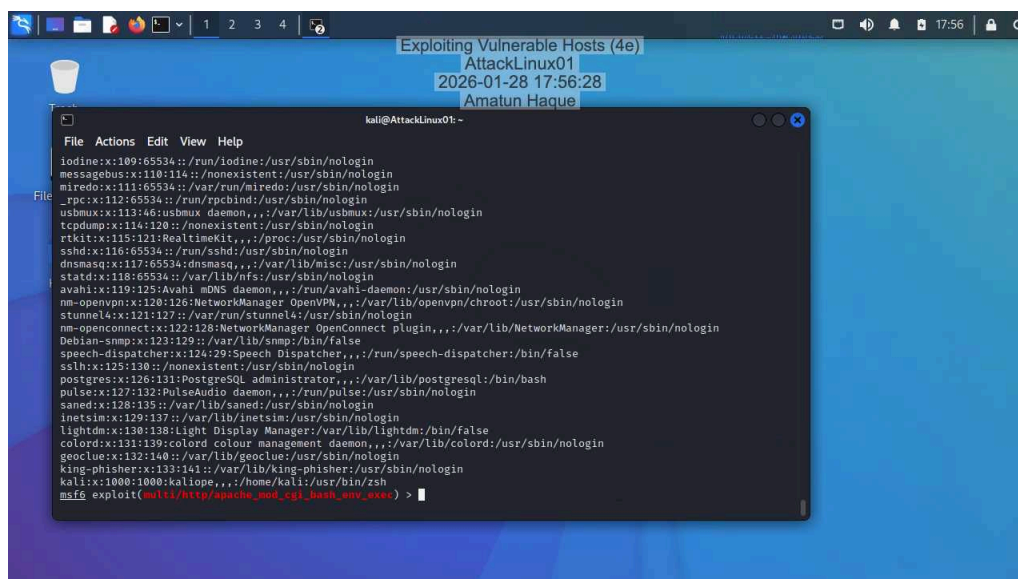
11. **Make a screen capture** showing the **login credentials** for the domain user on 172.30.0.20.



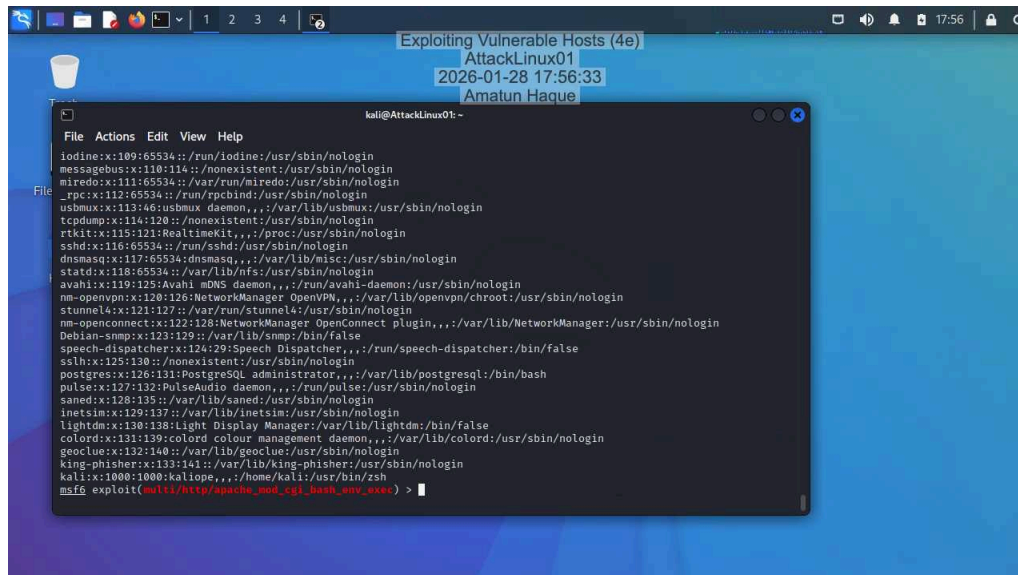
38. Make a screen capture showing the Meterpreter session for 172.30.0.30.



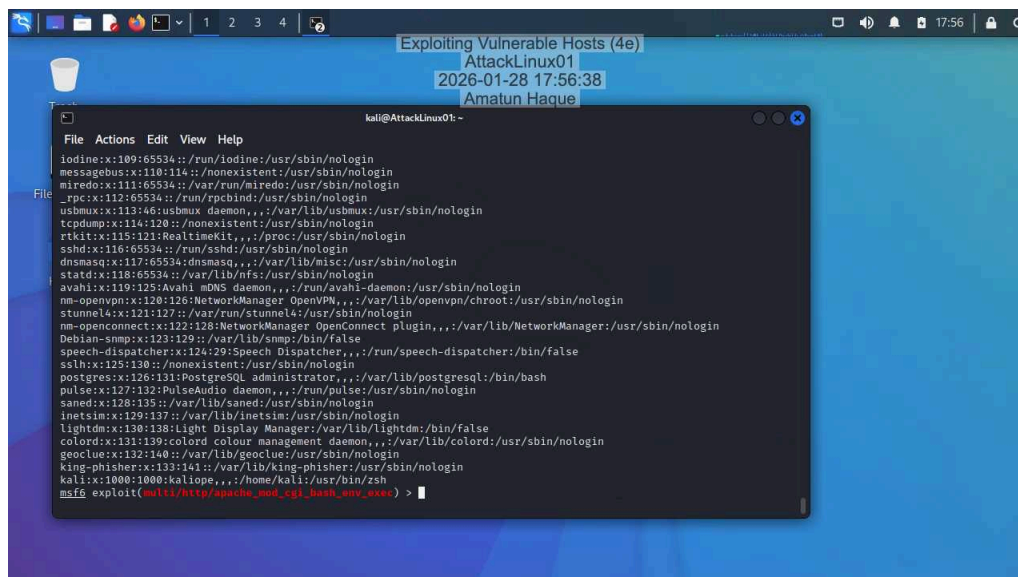
48. Make a screen capture showing the credentials found with `creds_wdigest`.



57. Make a screen capture showing the Meterpreter session for 172.30.0.40.

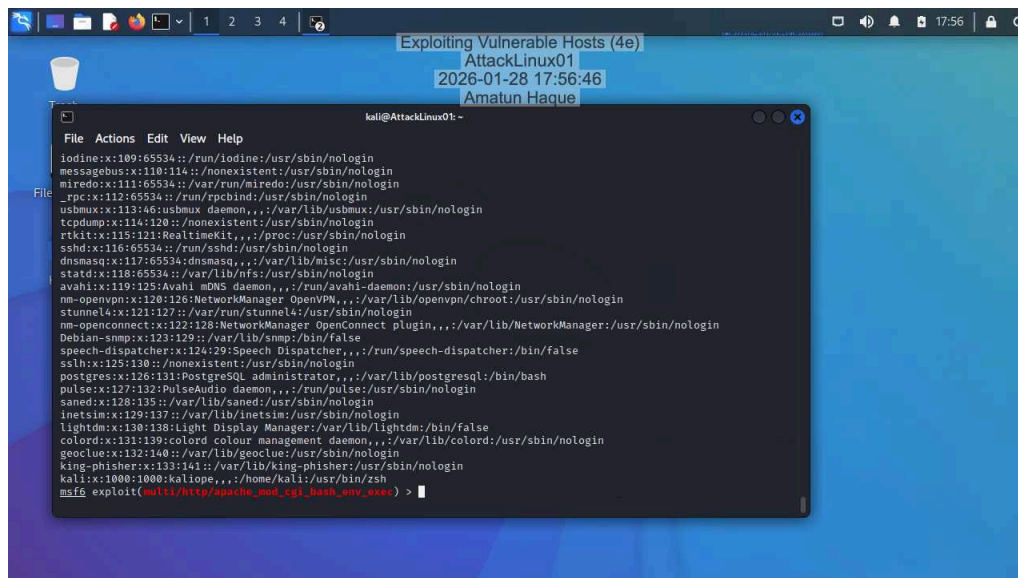


67. Make a screen capture showing the account information from 172.30.0.40.

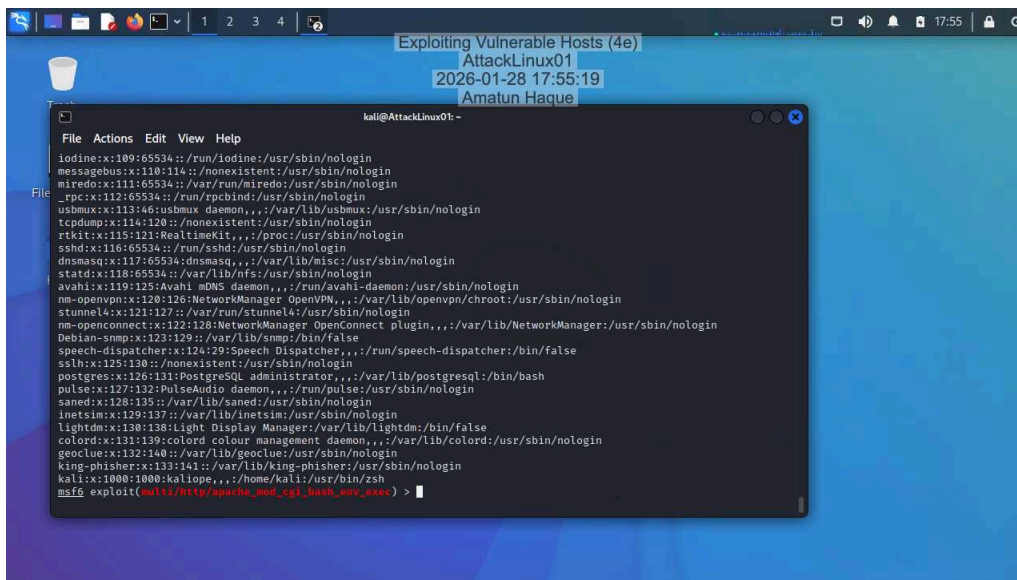


Part 3: Crack Password Hashes Using John the Ripper

5. Make a screen capture showing the user accounts and cracked passwords.



17. Make a screen capture showing the first 10 lines of the passwd file.



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal window has a title bar that reads "Exploiting Vulnerable Hosts (4e)" and "AttackLinux01". The terminal output shows the first 10 lines of the /etc/passwd file, which are:

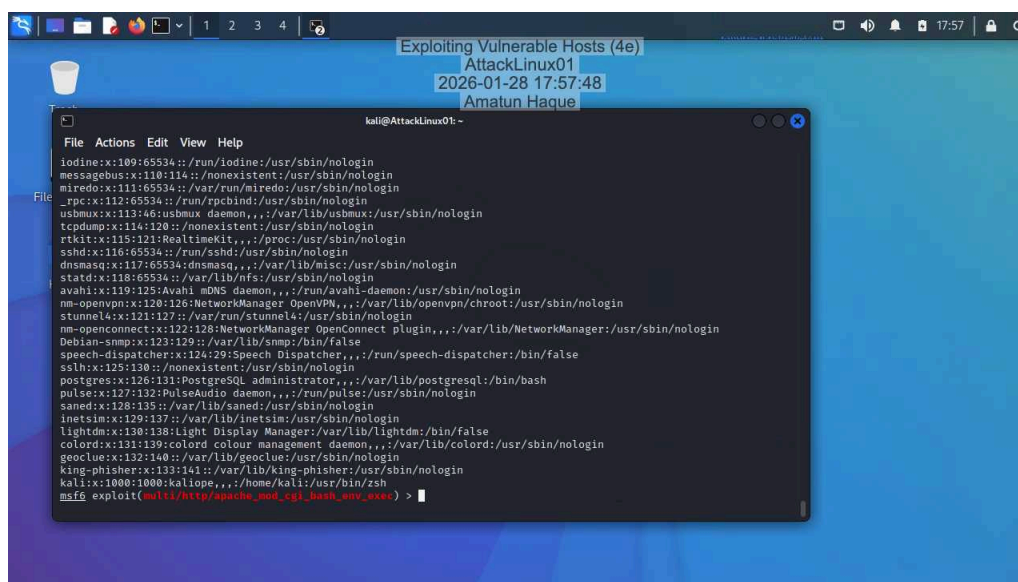
```
File Actions Edit View Help
iodine:x:109:65534::/run/iodine:/usr/sbin/nologin
messagebus:x:110:114::/nonexistent:/usr/sbin/nologin
miredo:x:111:65534::/var/run/miredo:/usr/sbin/nologin
_rpc:x:112:65534::/run/rpcbind:/usr/sbin/nologin
usbmux:x:113:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin
tcpdump:x:114:120::/nonexistent:/usr/sbin/nologin
rtkit:x:115:121:RealtimeKit,,:/proc:/usr/sbin/nologin
sshd:x:116:65534::/run/ssh:/usr/sbin/nologin
dnsmasq:x:117:65534:dnsmasq,,:/var/lib/misc:/usr/sbin/nologin
statd:x:118:65534::/var/lib/nfs:/usr/sbin/nologin
```

The terminal window also shows the prompt "kali@AttackLinux01: ~" and the command "cat /etc/passwd" being executed. The output of the command is displayed in the terminal window.

Section 3: Challenge and Analysis

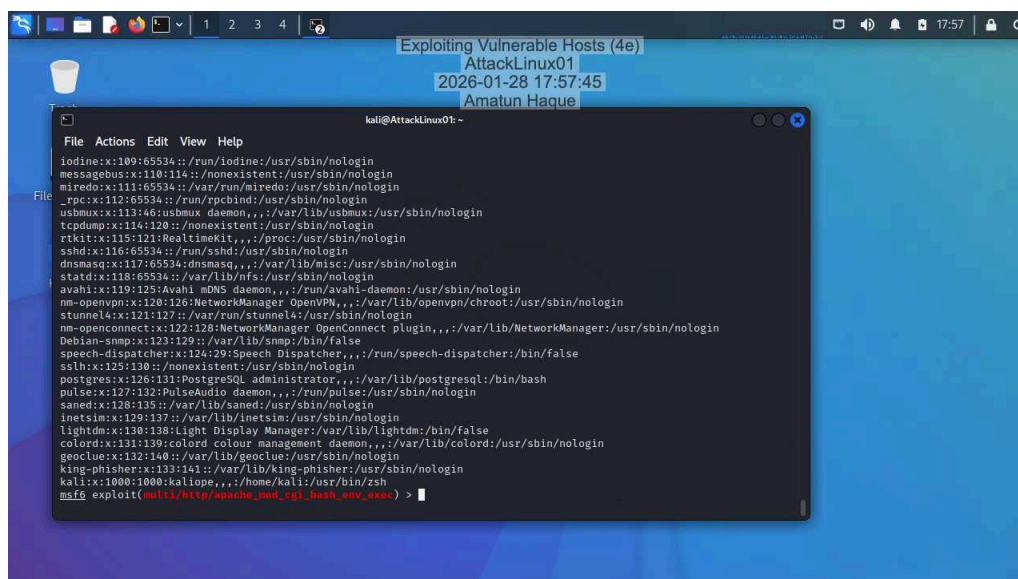
Part 1: Exploit the Domain Controller Directly

Make a screen capture showing a successful Meterpreter shell on the domain controller using this exploit.



Part 2: Get Hashes on a Linux System

Make a screen capture showing the hash for the root user.



Student:

Amatun Haque

Email:

amatunh.student@wust.edu

Time on Task:

25 hours, 18 minutes

Progress:

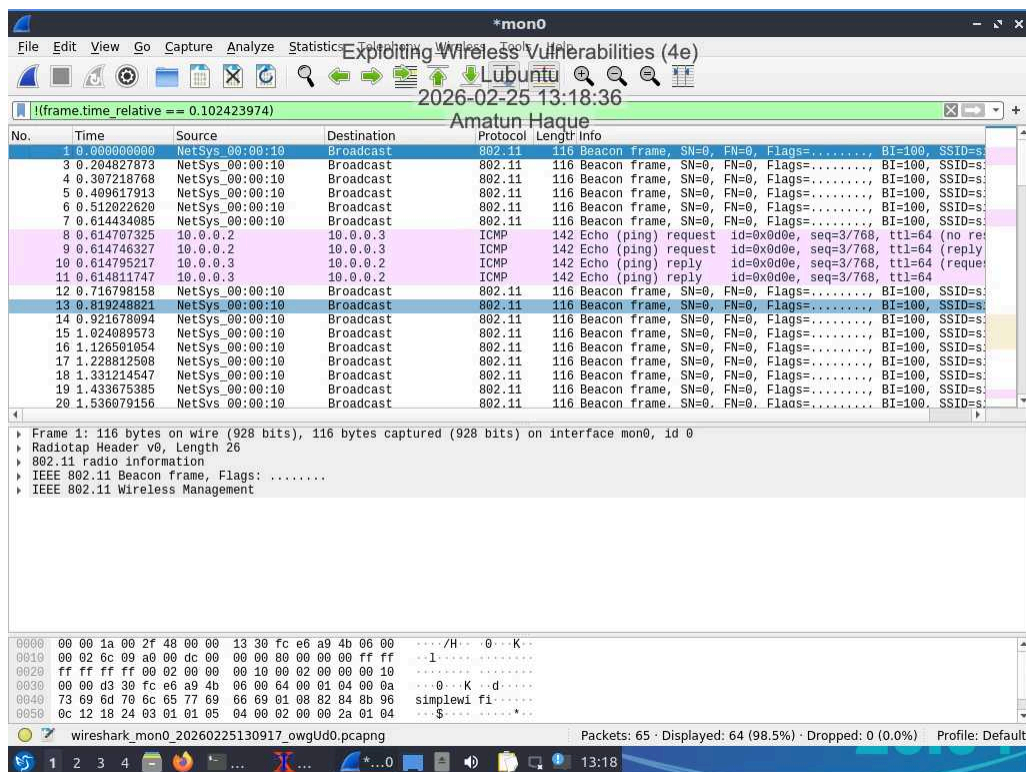
100%

Report Generated: Wednesday, February 25, 2026 at 2:03 PM

Section 1: Hands-On Demonstration

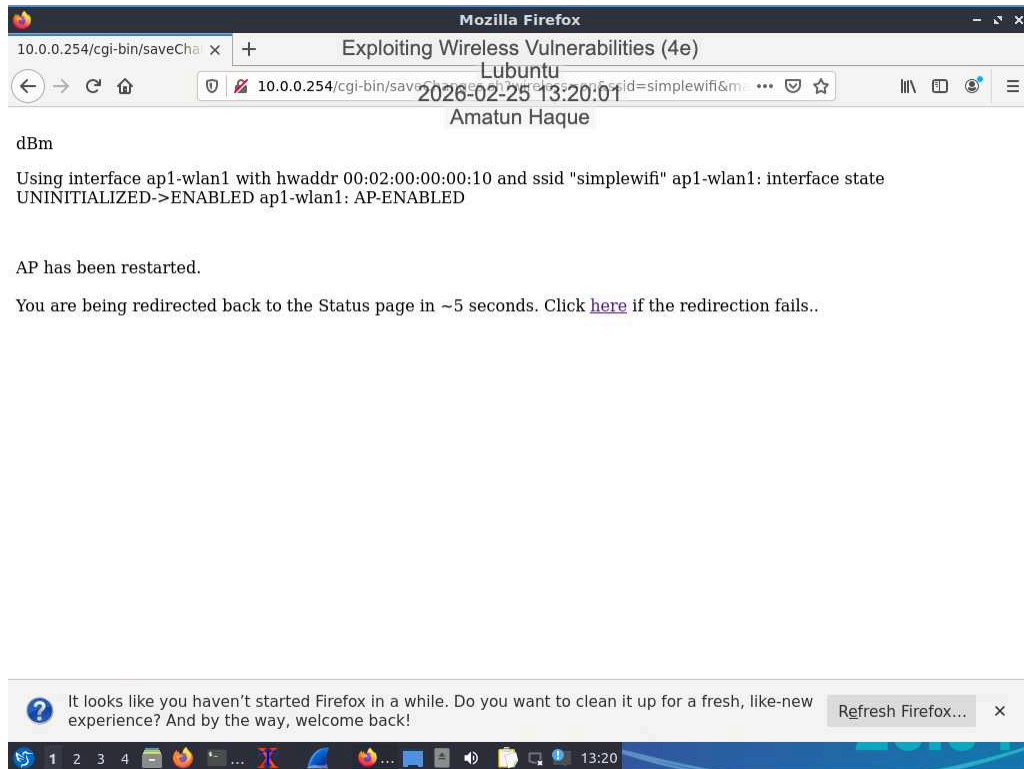
Part 1: Capture Unencrypted Traffic with Wireshark

16. Make a screen capture showing the HTTP headers in the Packet Bytes pane.

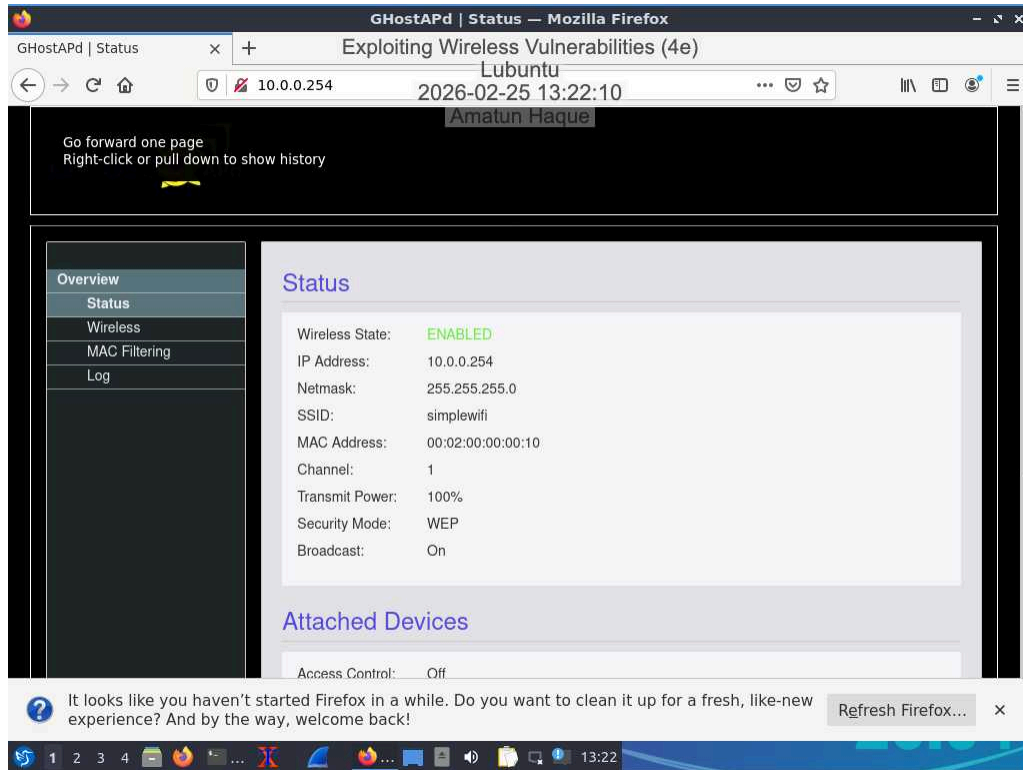


Part 2: Encrypt Wireless Traffic with WEP

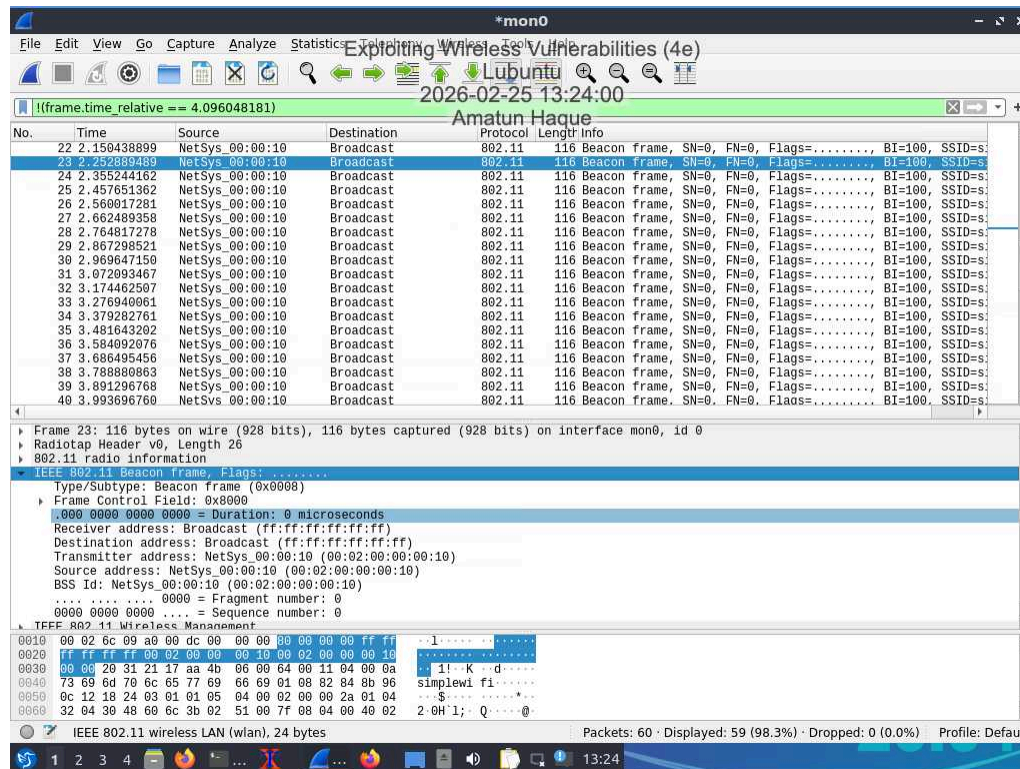
7. Make a screen capture showing WEP mode enabled on the GHostAPd Status page.



14. Make a screen capture showing WEP mode enabled and both sta2 and sta3 devices attached on the GHostAPd Status page.

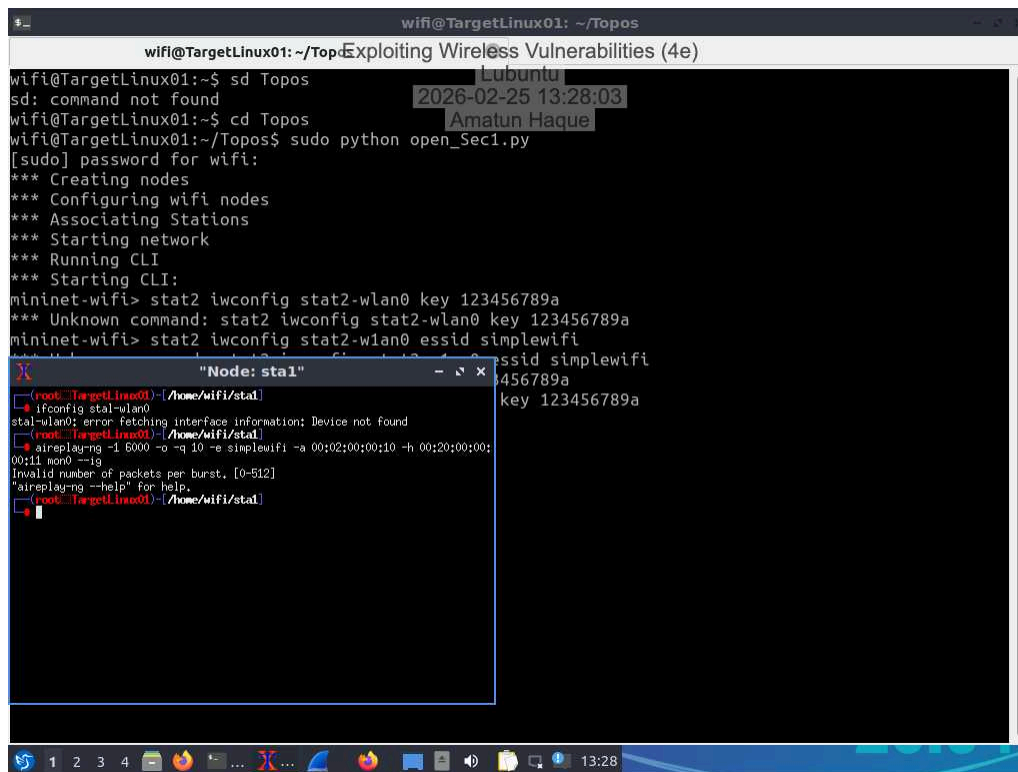


24. Make a screen capture showing the Initialization Vector value in the Packet Details pane.



Part 3: Break WEP Encryption

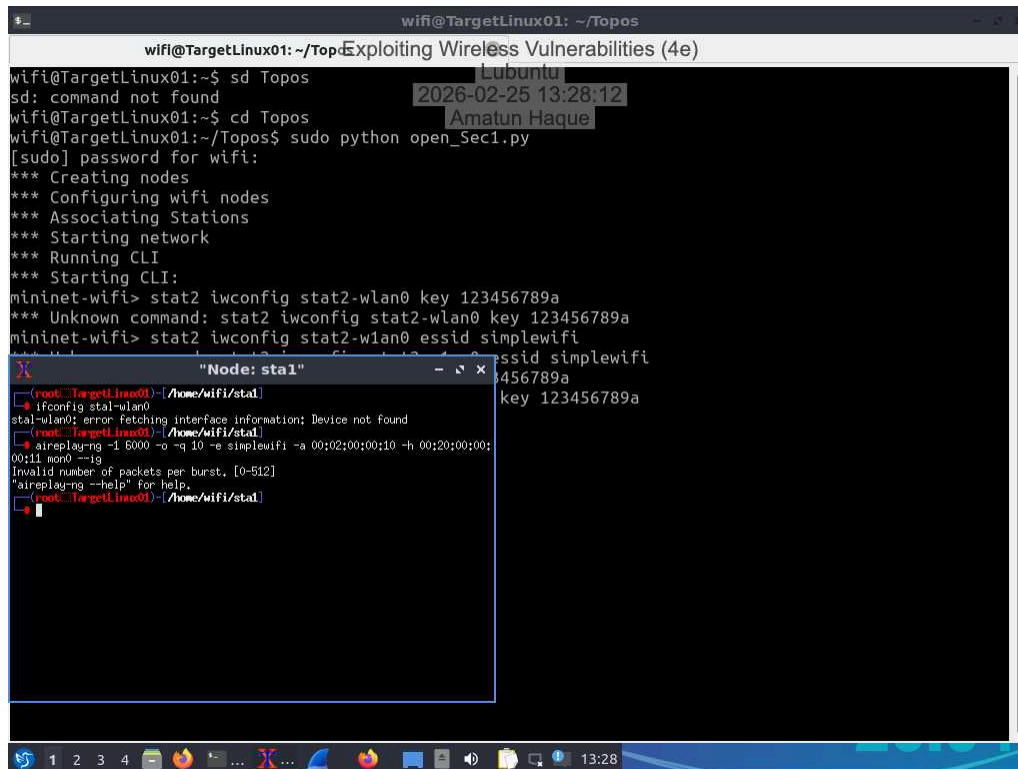
14. Make a screen capture showing **KEY FOUND** in your aircrack-ng output.



```
wifi@TargetLinux01: ~/Topos
wifi@TargetLinux01:~$ sd Topos
sd: command not found
wifi@TargetLinux01:~$ cd Topos
wifi@TargetLinux01:~/Topos$ sudo python open_Sec1.py
[sudo] password for wifi:
*** Creating nodes
*** Configuring wifi nodes
*** Associating Stations
*** Starting network
*** Running CLI
*** Starting CLI:
mininet-wifi> stat2 iwconfig stat2-wlan0 key 123456789a
*** Unknown command: stat2 iwconfig stat2-wlan0 key 123456789a
mininet-wifi> stat2 iwconfig stat2-wlan0 essid simplewifi
*** Unknown command: stat2 iwconfig stat2-wlan0 essid simplewifi
*** Unknown command: stat2 iwconfig stat2-wlan0 key 123456789a
*** Unknown command: stat2 iwconfig stat2-wlan0 key 123456789a

Node: sta1
(root@TargetLinux01)~/home/wifi/sta1
ifconfig stat2-wlan0
stat2-wlan0: error fetching interface information: Device not found
(root@TargetLinux01)~/home/wifi/sta1
aireplay-ng -i 6000 -o -q 10 -e simplewifi -a 00:02:00:00:10 -h 00:20:00:00:
00:11 mon0 --ig
Invalid number of packets per burst, [0-512]
"aireplay-ng --help" for help.
(root@TargetLinux01)~/home/wifi/sta1
```

27. Make a screen capture showing the decrypted Hypertext Transfer Protocol data.



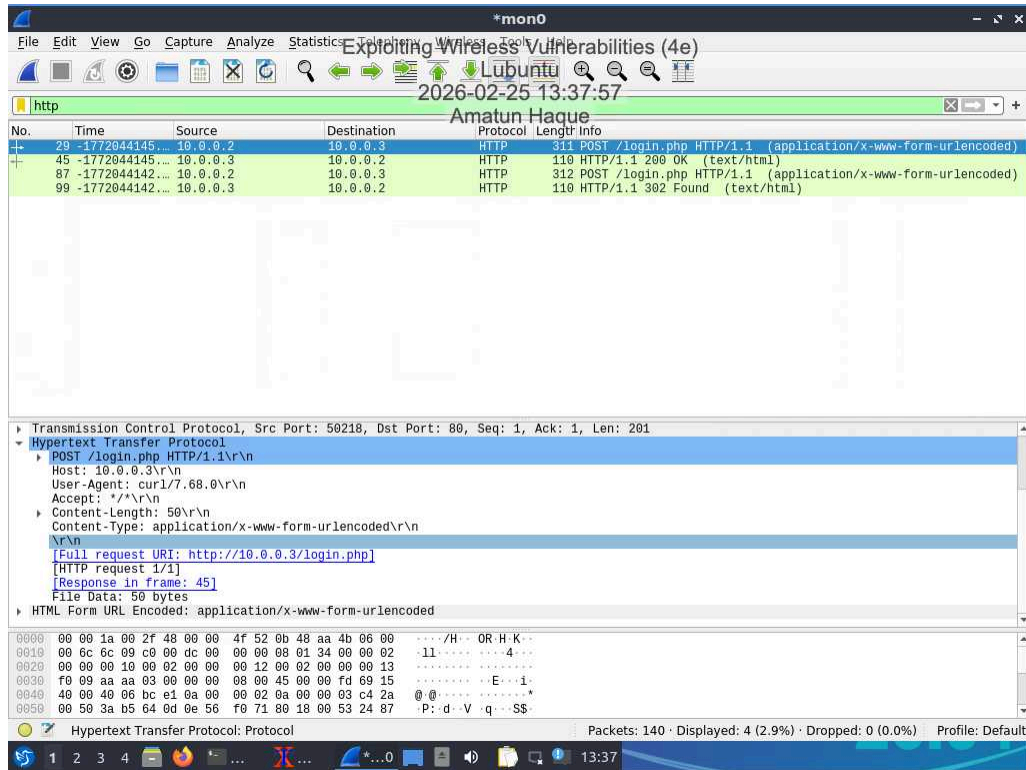
```
wifi@TargetLinux01: ~/Topos
wifi@TargetLinux01:~/Topos$ sd Topos
sd: command not found
wifi@TargetLinux01:~/Topos$ cd Topos
wifi@TargetLinux01:~/Topos$ sudo python open_Sec1.py
[sudo] password for wifi:
*** Creating nodes
*** Configuring wifi nodes
*** Associating Stations
*** Starting network
*** Running CLI
*** Starting CLI:
mininet-wifi> stat2 iwconfig stat2-wlan0 key 123456789a
*** Unknown command: stat2 iwconfig stat2-wlan0 key 123456789a
mininet-wifi> stat2 iwconfig stat2-wlan0 essid simplewifi
mininet-wifi> stat2 iwconfig stat2-wlan0 key 123456789a
mininet-wifi> stat2 iwconfig stat2-wlan0 key 123456789a

"Node: sta1"
(root@TargetLinux01) [/home/wifi/sta1]
# ifconfig sta1-wlan0
sta1-wlan0: error fetching interface information: Device not found
(root@TargetLinux01) [/home/wifi/sta1]
# aireplay-ng -i 6000 -o -q 10 -e simplewifi -a 00:02:00:00:10 -h 00:20:00:00:00:11 -m 00:11:00:00:11
Invalid number of packets per burst. [0-512]
"aireplay-ng --help" for help.
(root@TargetLinux01) [/home/wifi/sta1]
```


Section 2: Applied Learning

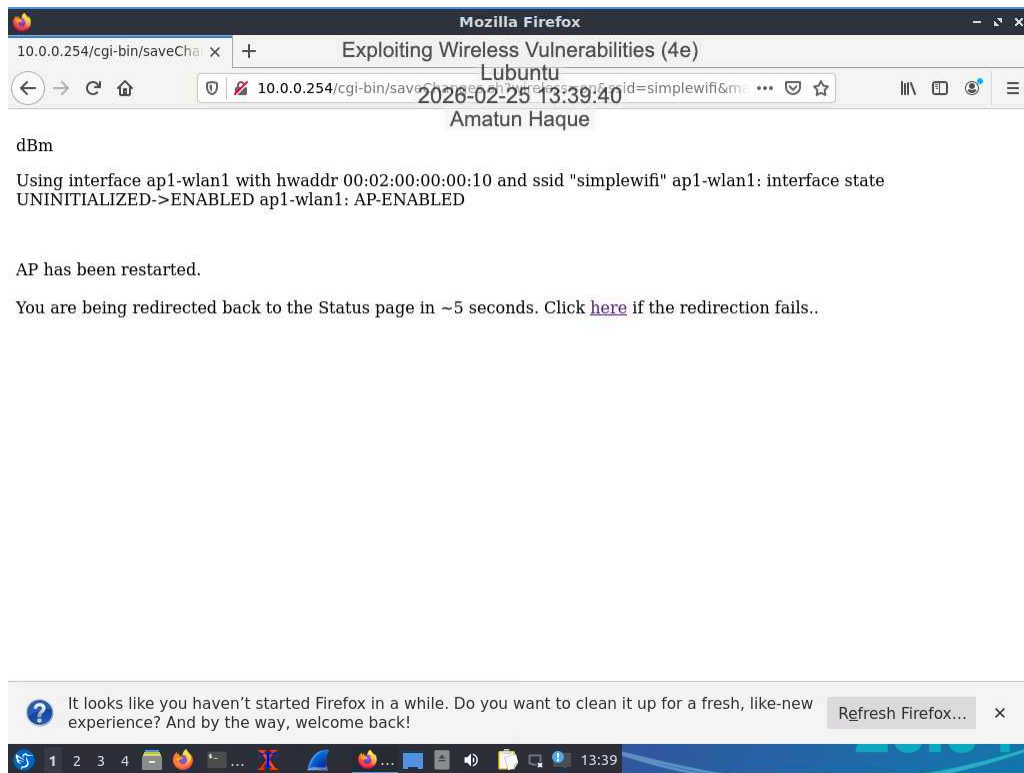
Part 1: Capture Unencrypted Traffic with Wireshark

15. Make a screen capture showing the “Username” and “Password” form items in the Packet Details pane.

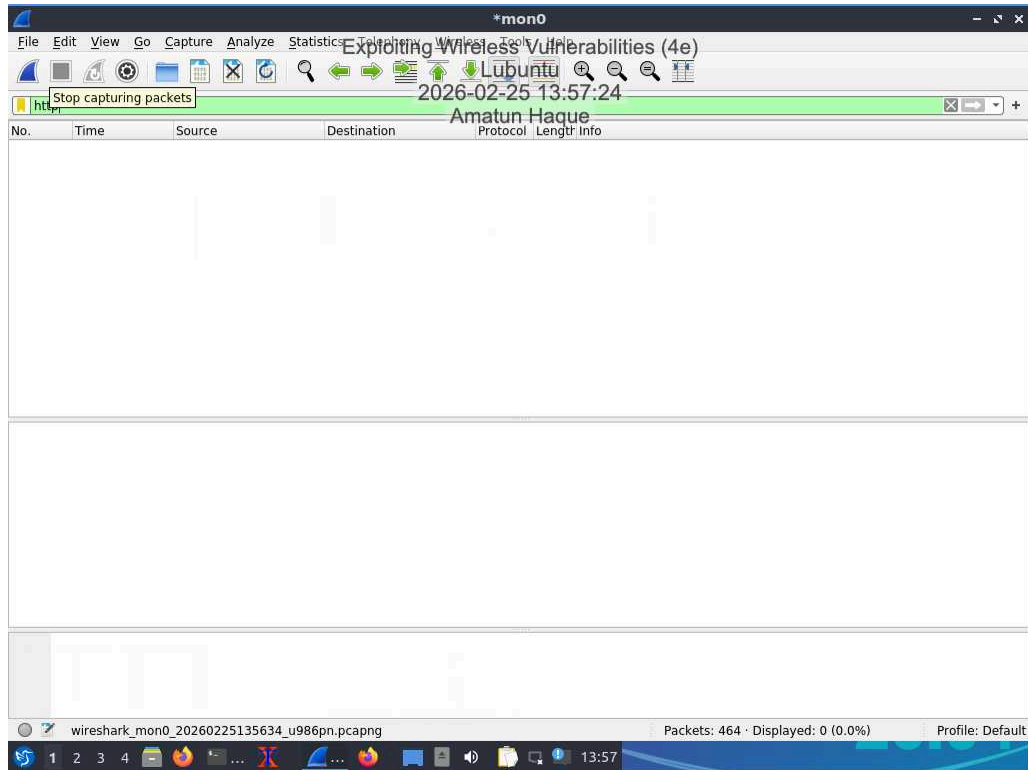


Part 2: Encrypt Wireless Traffic with WPA2

6. Make a screen capture showing the **GHostAPd Status** page with WPA2 enabled as the Security Mode.

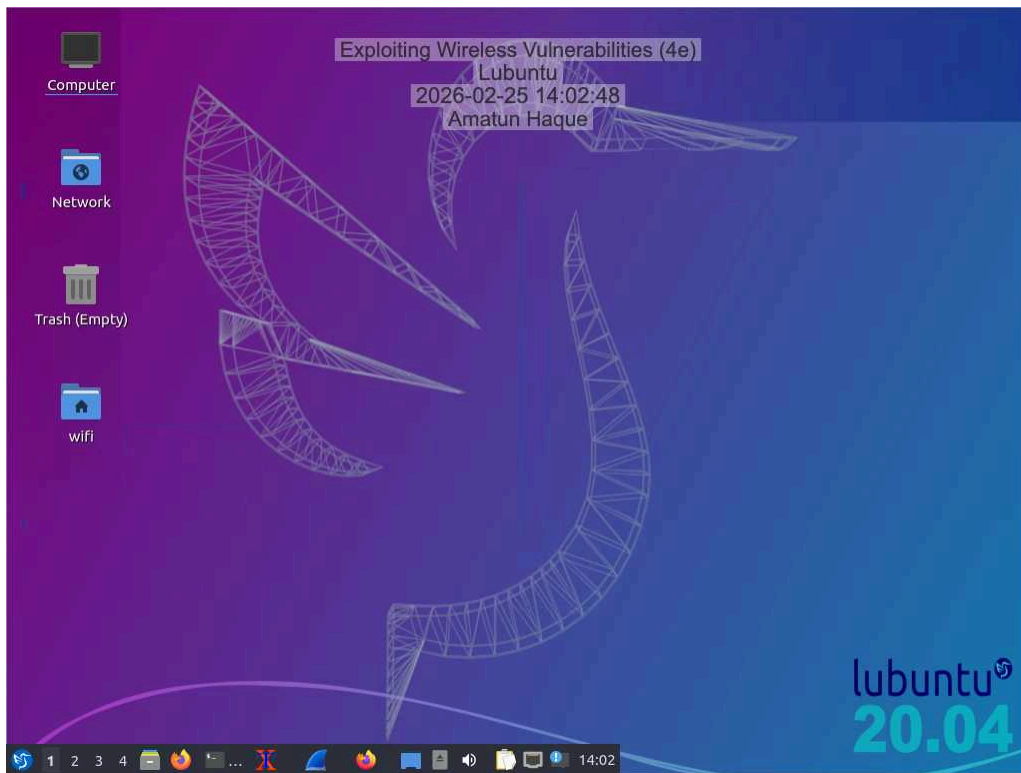


21. Make a screen capture showing the **CCMP Ext. Initialization Vector** in the **Packet Details** pane.

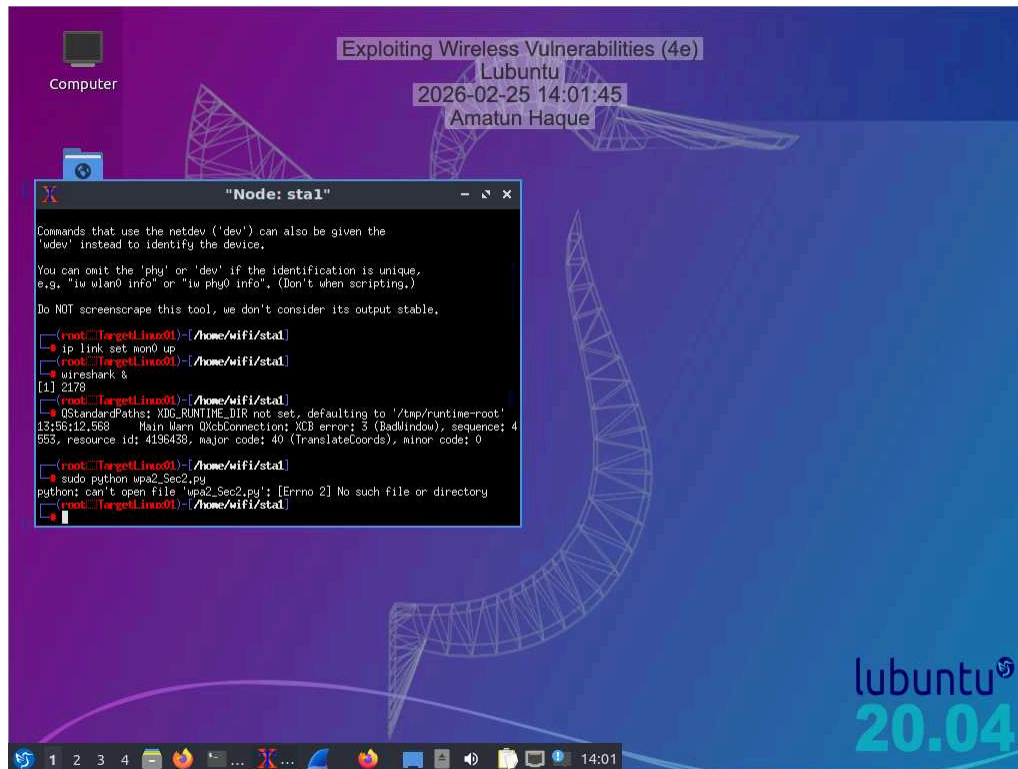


Part 3: Break WPA2 Encryption

21. **Make a screen capture** showing the **length of your new *yourname_Capture.txt* wordlist in the JtR output.**



23. Make a screen capture showing the **discovered passphrase** in your aircrack output.



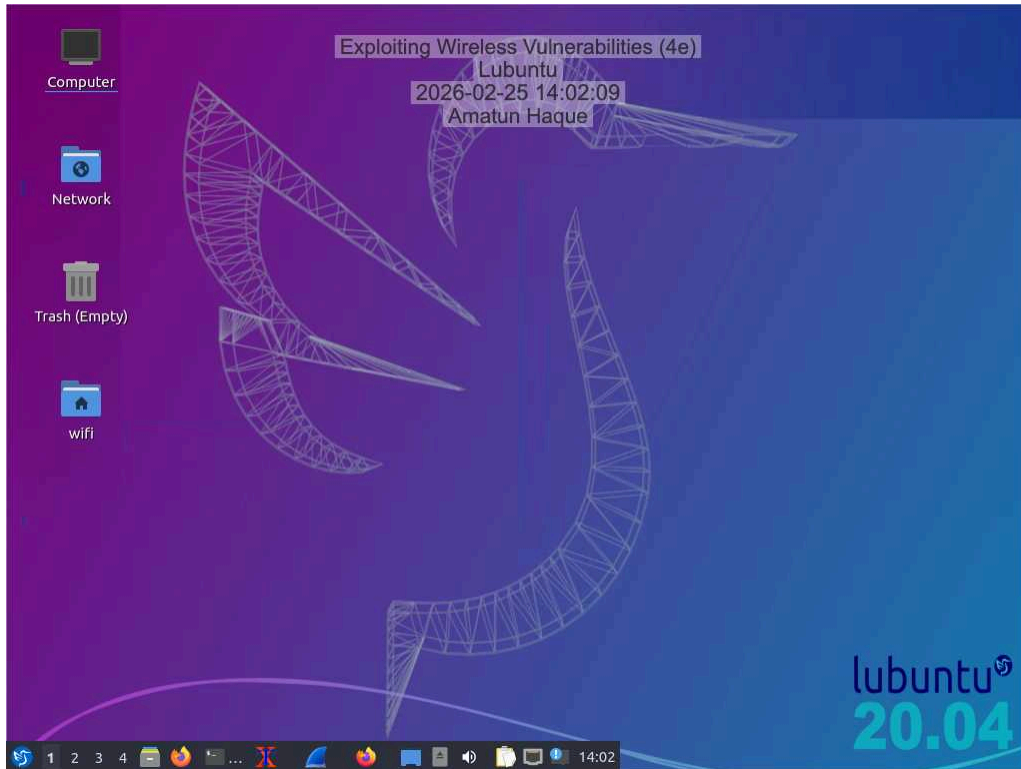
32. Record the **password** discovered for the FTP user in your Wireshark packet capture.

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Section 3: Challenge and Analysis

Part 1: Mangle a Wordlist with John the Ripper

Make a screen capture showing the output from your john command used to generate rodRage_Final.lst.

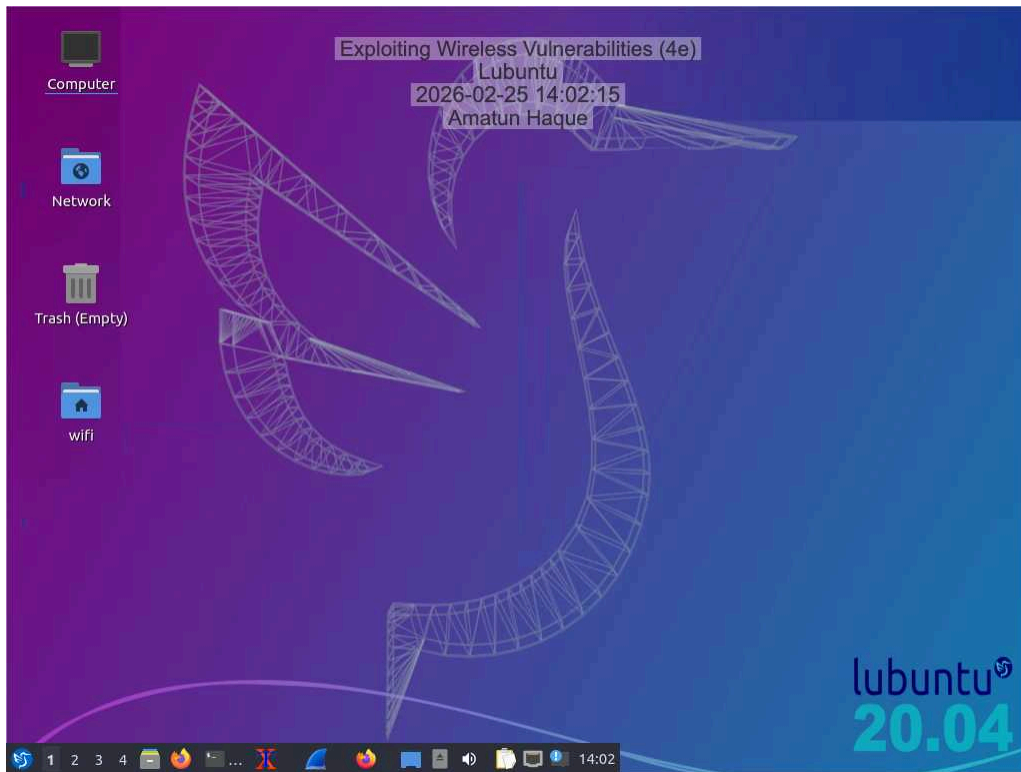


Part 2: Perform a Dictionary Attack using a WPA2 Network Capture

Exploiting Wireless Vulnerabilities (4e)

Ethical Hacking, Fourth Edition - Lab 07

Make a screen capture showing the **recovered WPA2 passphrase** in your **aircrack-ng** output.



Investigating and Responding to Security Incidents (4e)

Ethical Hacking, Fourth Edition - Lab 09

Student:

Amatun Haque

Email:

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Time on Task:

24 hours, 42 minutes

Progress:

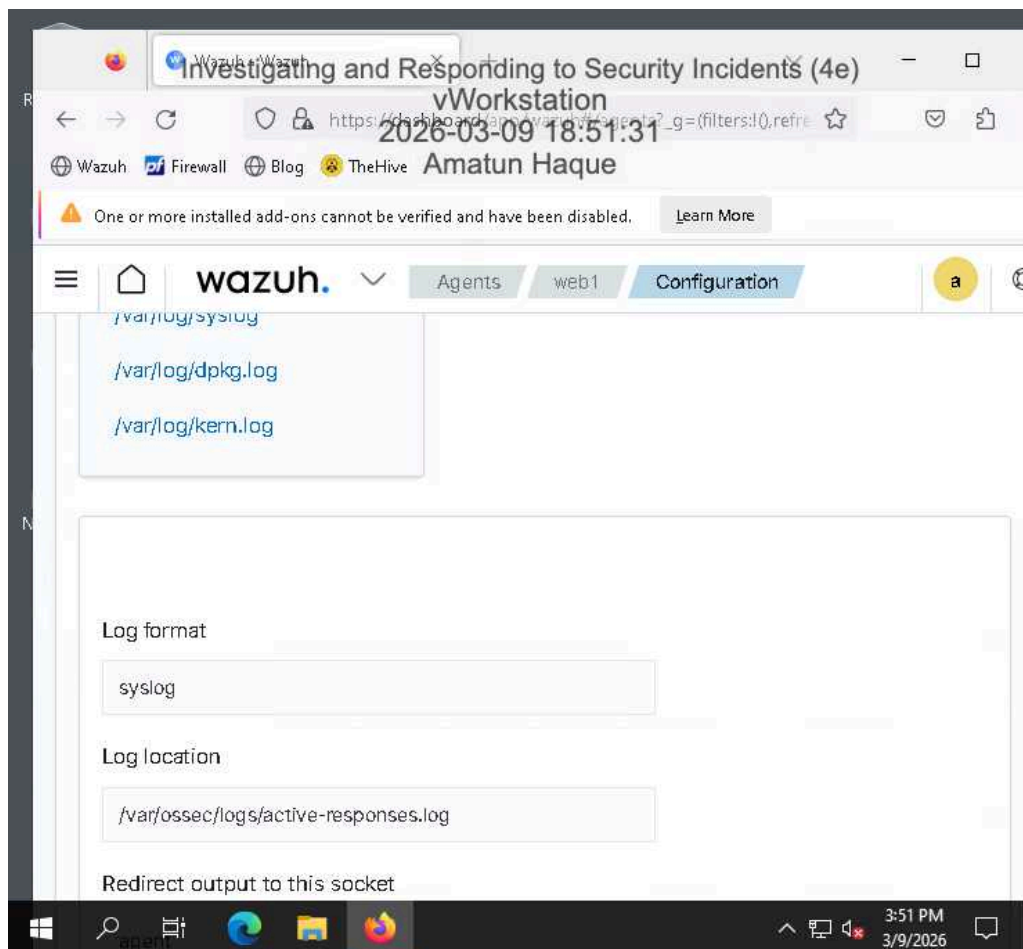
100%

Report Generated: Wednesday, March 11, 2026 at 4:37 PM

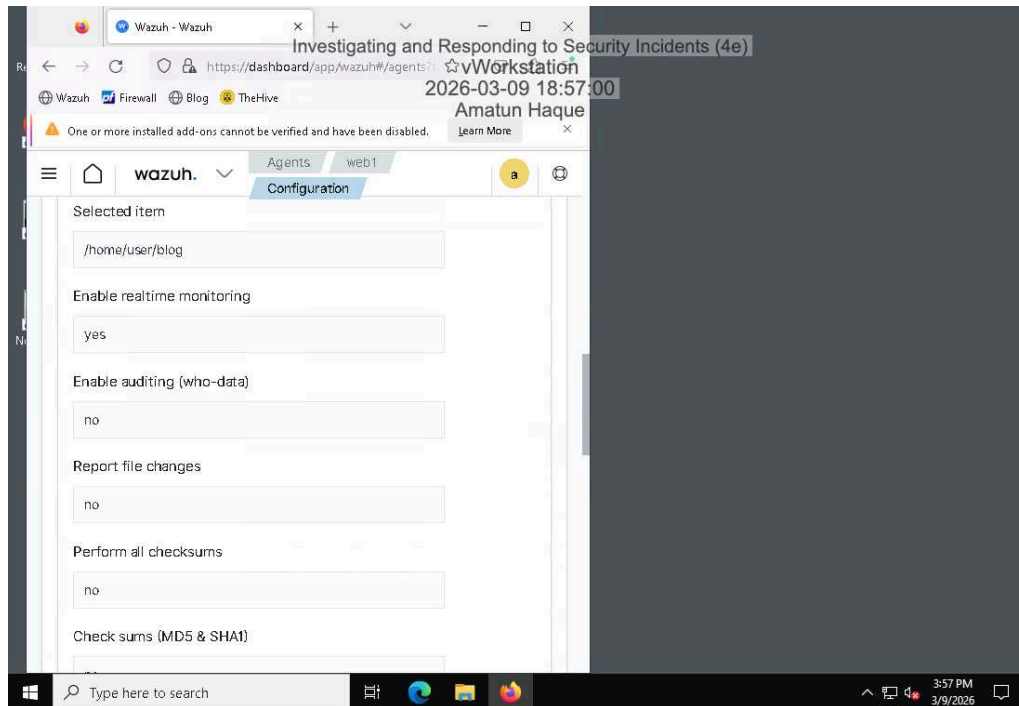
Hands-On Demonstration

Part 1: Review the SIEM configuration

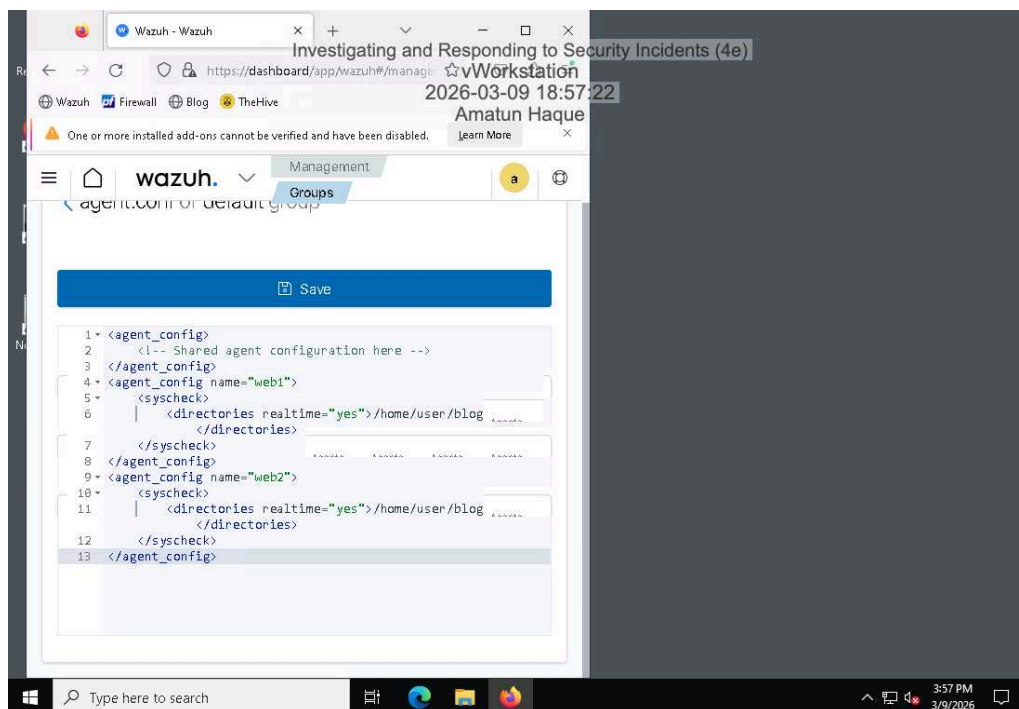
8. Make a screen capture showing the list of log files monitored on web1.



13. Make a screen capture showing the real time monitoring setting for the blog folder.

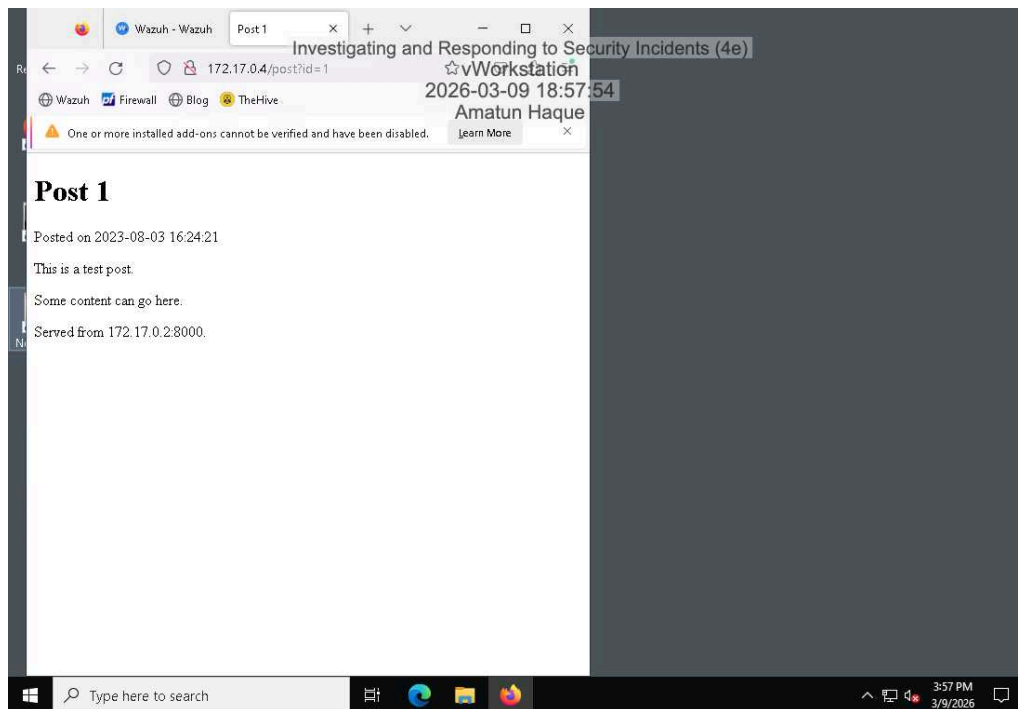


16. Make a screen capture showing the default group configuration.

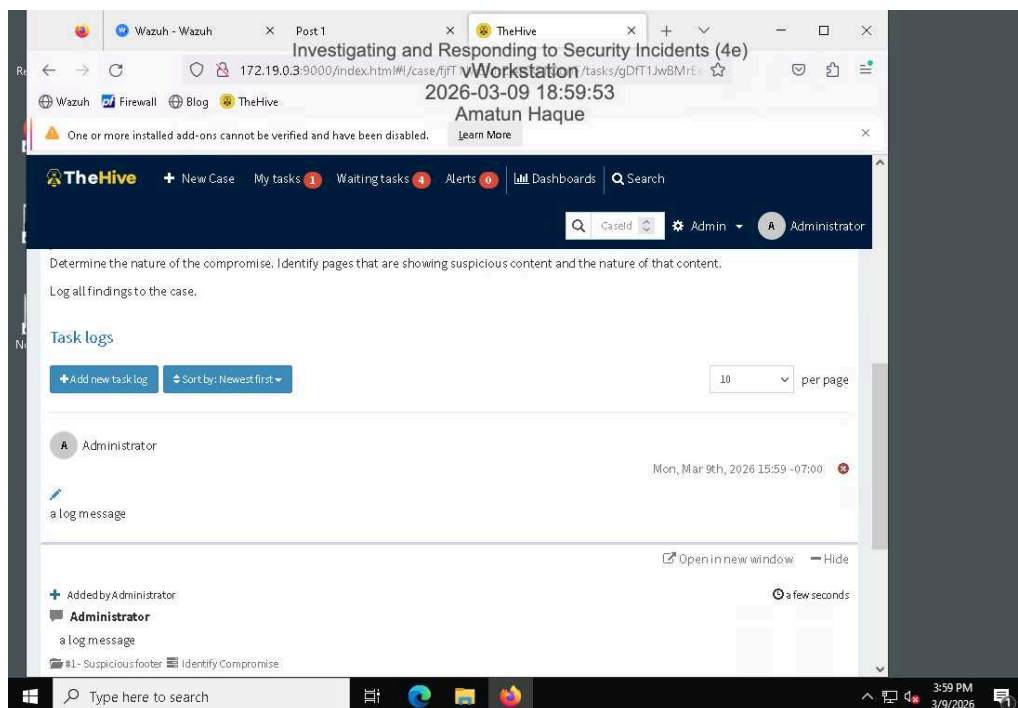


Part 2: Identify the scope of a reported issue

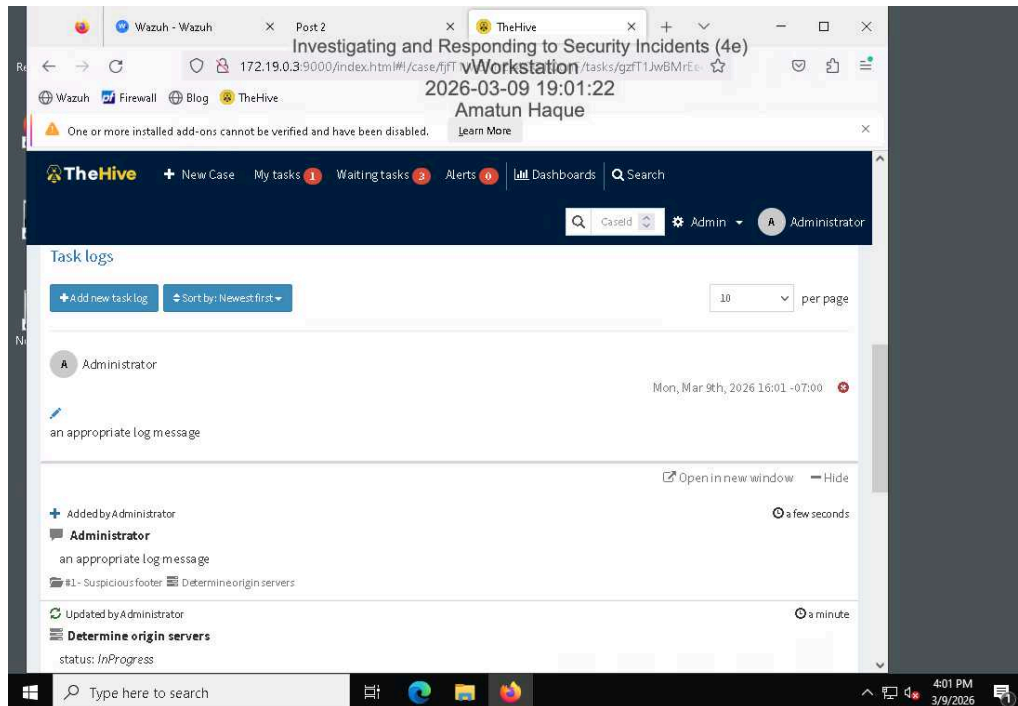
6. **Make a screen capture** showing the blog post with the suspicious content.



17. **Make a screen capture** showing the log entry in the Identify compromise tab.

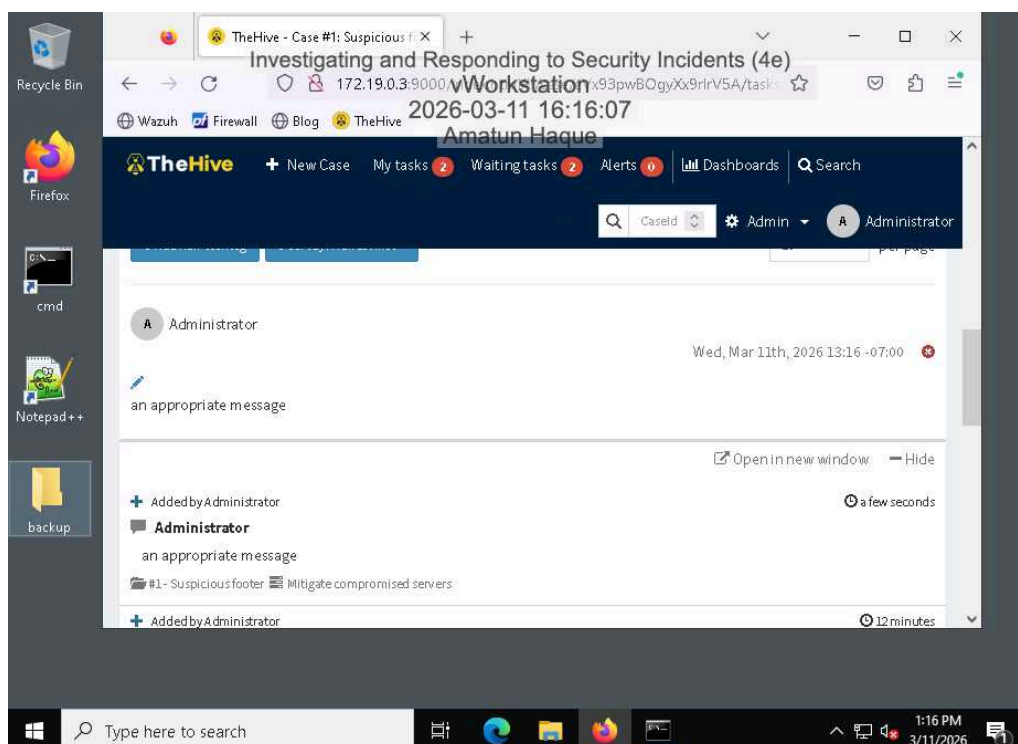


27. Make a screen capture showing the Determine origin servers log entry.

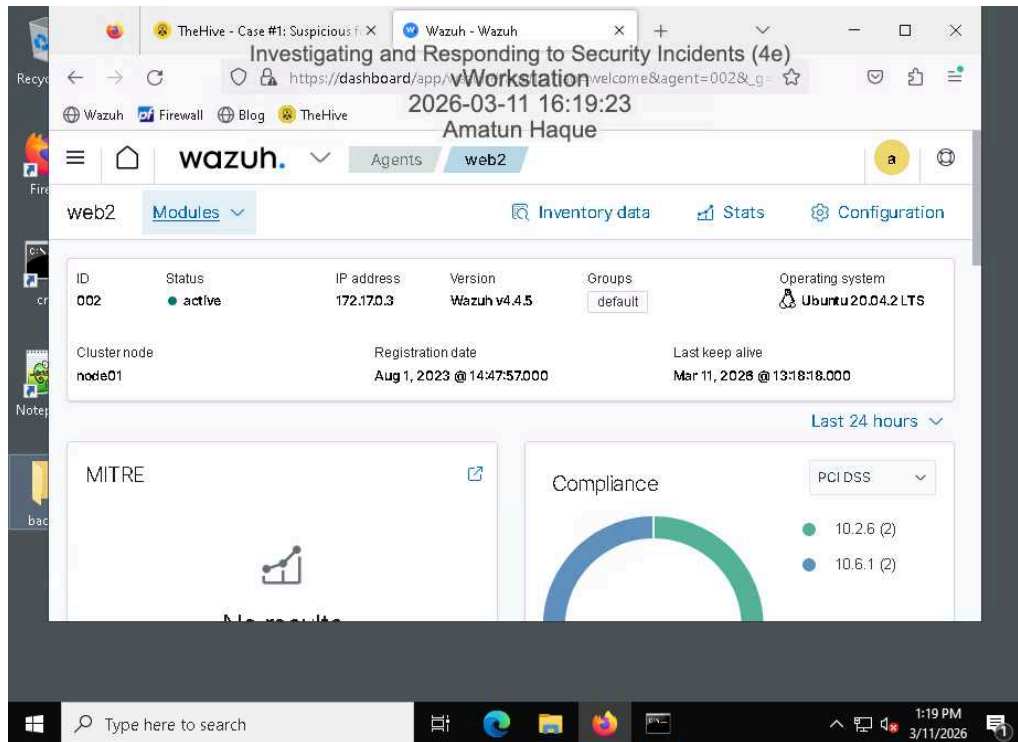


Part 3: Contain and Investigate the event

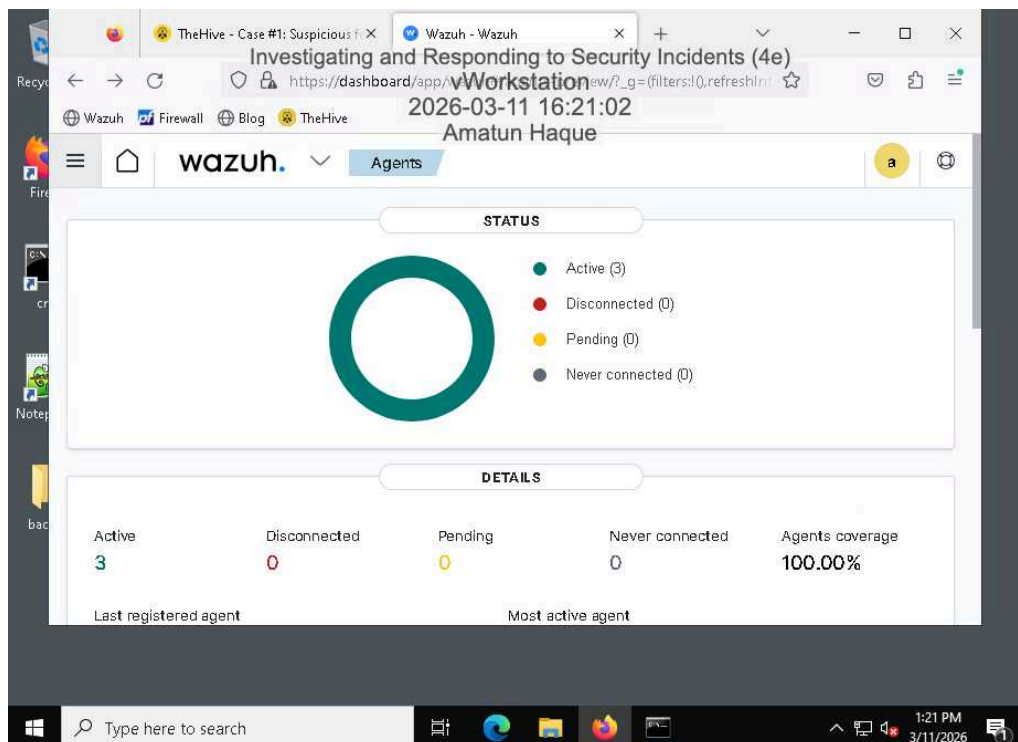
16. Make a screen capture showing the log entry for the Mitigate compromised servers task.



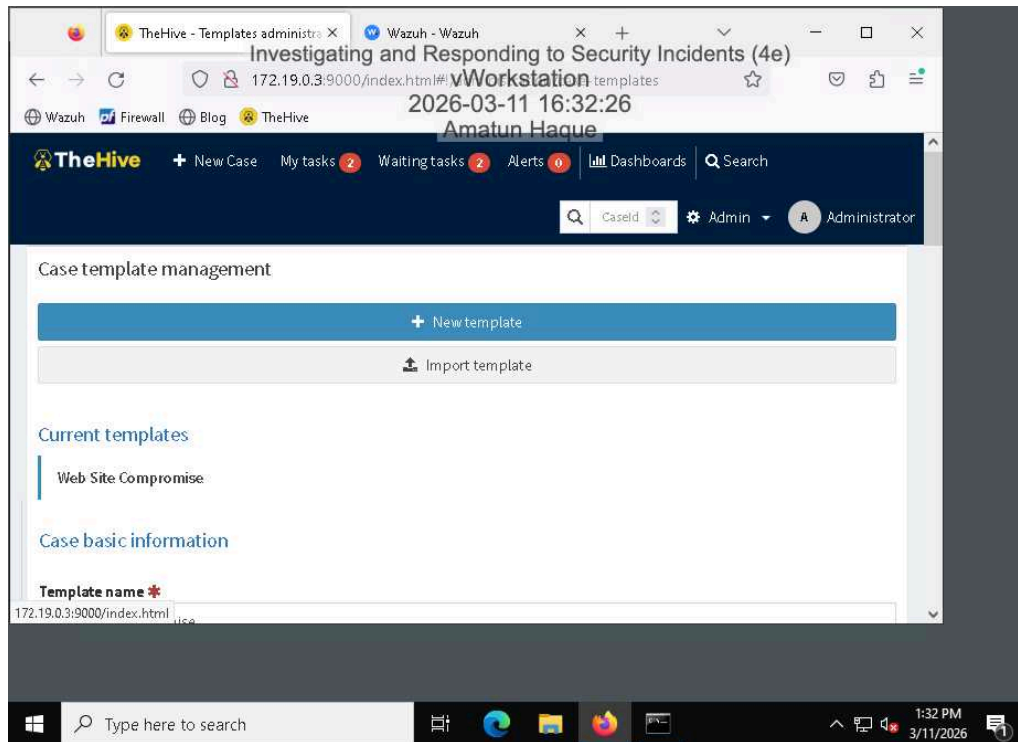
24. Make a screen capture showing at least the first three events in the list.



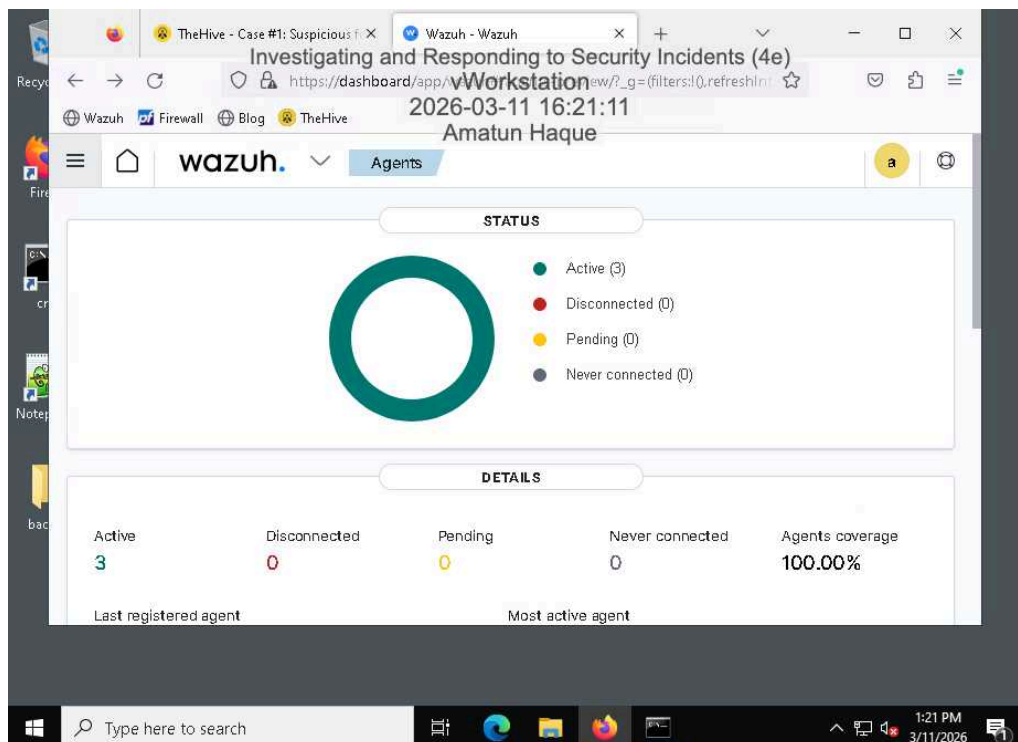
34. Make a screen capture showing the full_log field from the SSH event.



38. Make a screen capture showing the list of key fingerprints.

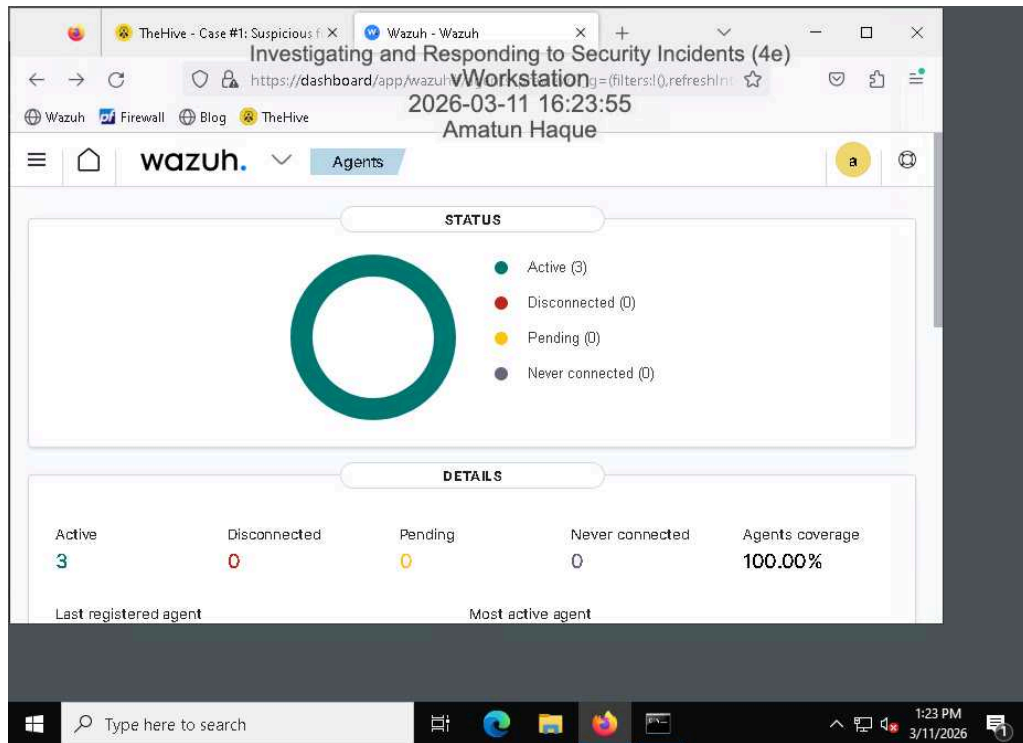


47. Make a screen capture showing the log entry for the Determine the root cause task.



Part 4: Recover from the incident

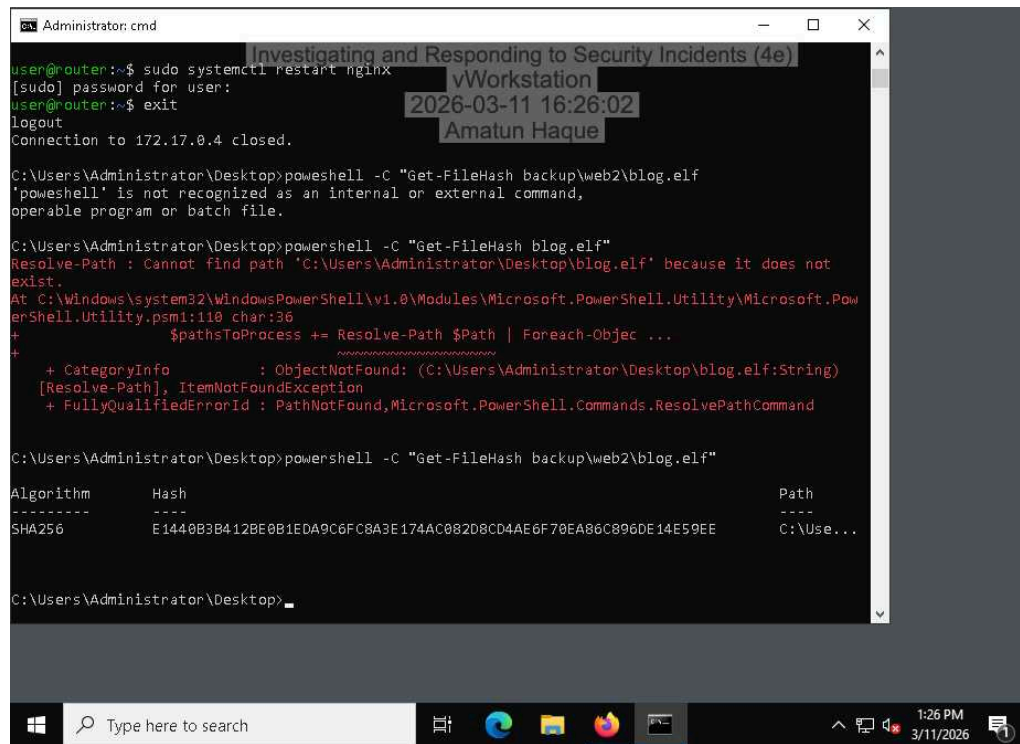
13. Make a screen capture showing the log entry for the Restore Servers task.



Applied Learning

Part 1: Investigate the root cause using log files

4. Make a screen capture showing the different file hashes



```
Administrator: cmd
user@router:~$ sudo systemctl restart nginx
[sudo] password for user:
user@router:~$ exit
logout
Connection to 172.17.0.4 closed.

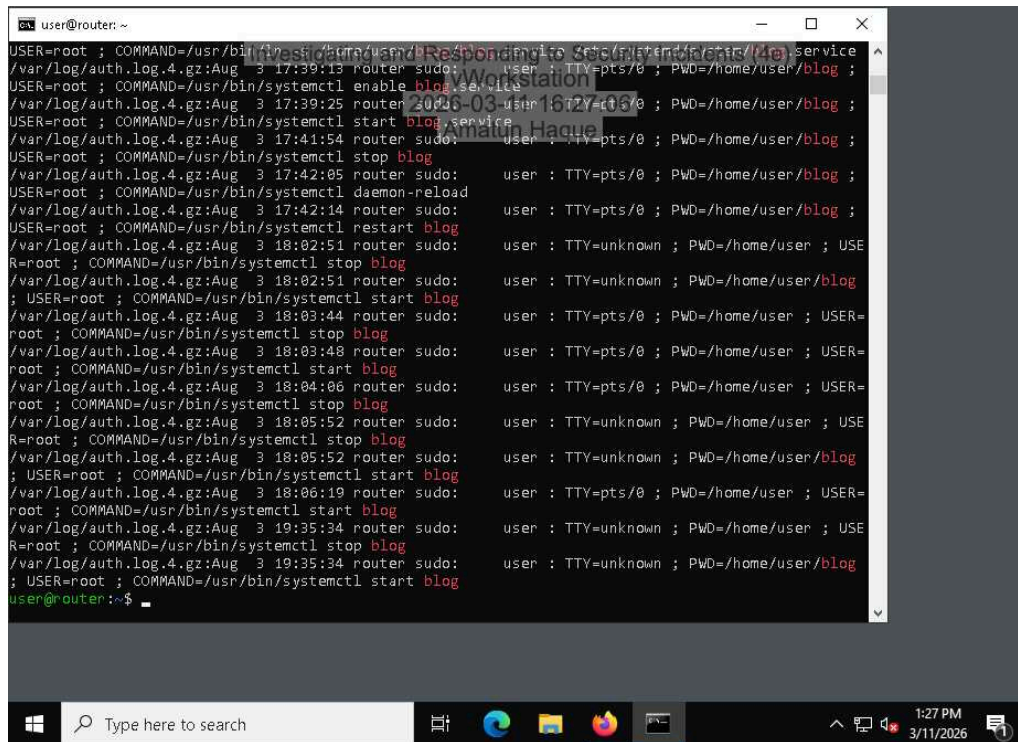
C:\Users\Administrator\Desktop>powershell -C "Get-FileHash backup\web2\blog.elf"
'powershell' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator\Desktop>powershell -C "Get-FileHash blog.elf"
Resolve-Path : Cannot find path 'C:\Users\Administrator\Desktop\blog.elf' because it does not
exist.
At C:\Windows\system32\WindowsPowerShell\v1.0\Modules\Microsoft.PowerShell.Utility\Microsoft.Pow
erShell.Utility.psm1:118 char:36
+ $pathsToProcess += Resolve-Path $Path | Foreach-Objec ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\Administrator\Desktop\blog.elf:String)
[Resolve-Path], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.ResolvePathCommand

C:\Users\Administrator\Desktop>powershell -C "Get-FileHash backup\web2\blog.elf"

Algorithm      Hash                                          Path
-----
SHA256         E1440B3B412BE0B1EDA9C6FC8A3E174AC082D8CD4AE6F70EA86C896DE14E59EE C:\Use...
```

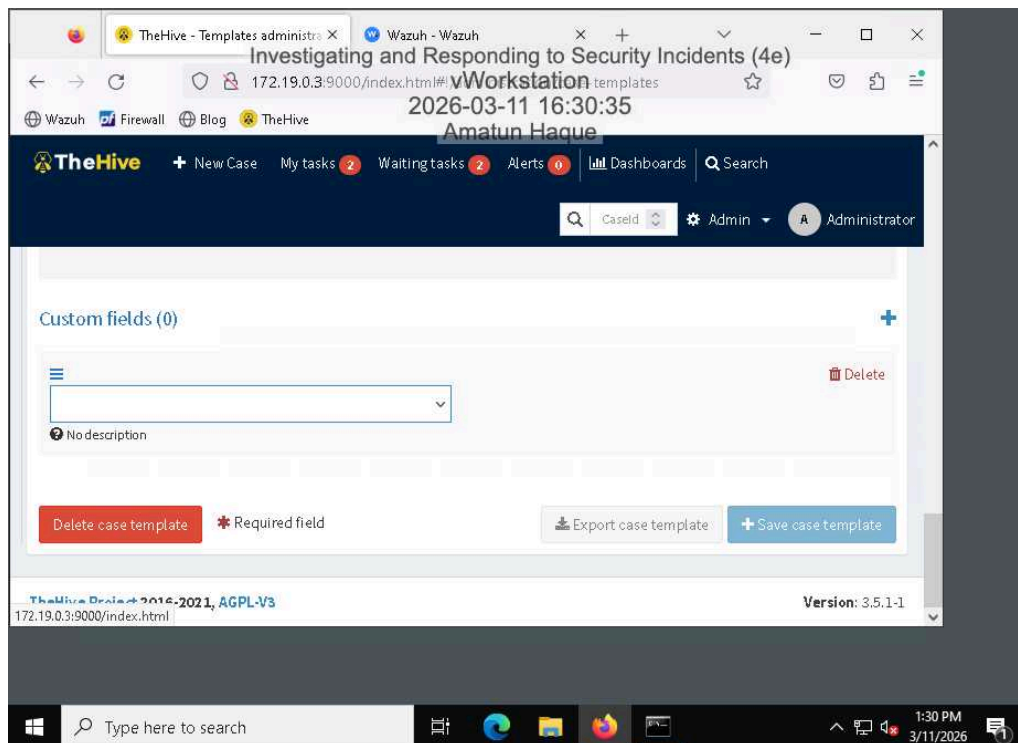
8. Make a screen capture showing all the events found using the previous two commands.



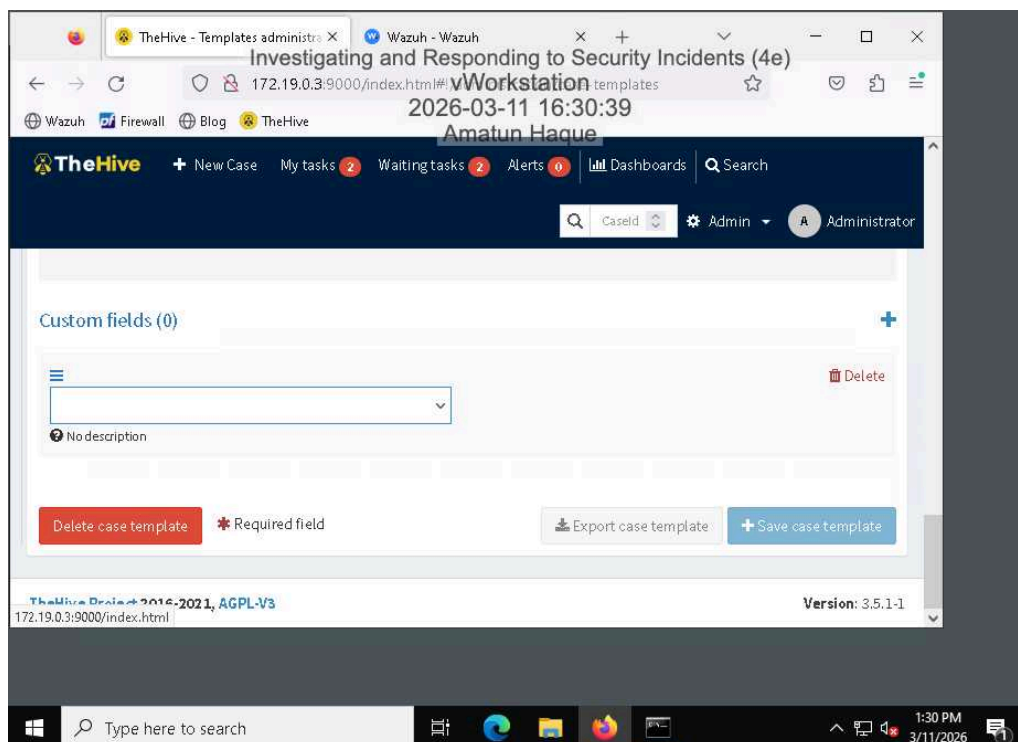
```
user@router: ~  
USER=root ; COMMAND=/usr/bin/systemctl enable blog.service  
/var/log/auth.log.4.gz:Aug  3 17:39:13 router sudo: user : TTY=pts/0 ; PWD=/home/user/blog ;  
USER=root ; COMMAND=/usr/bin/systemctl enable blog.service  
/var/log/auth.log.4.gz:Aug  3 17:39:25 router sudo: user : TTY=pts/0 ; PWD=/home/user/blog ;  
USER=root ; COMMAND=/usr/bin/systemctl start blog.service  
/var/log/auth.log.4.gz:Aug  3 17:41:54 router sudo: user : TTY=pts/0 ; PWD=/home/user/blog ;  
USER=root ; COMMAND=/usr/bin/systemctl stop blog  
/var/log/auth.log.4.gz:Aug  3 17:42:05 router sudo: user : TTY=pts/0 ; PWD=/home/user/blog ;  
USER=root ; COMMAND=/usr/bin/systemctl daemon-reload  
/var/log/auth.log.4.gz:Aug  3 17:42:14 router sudo: user : TTY=pts/0 ; PWD=/home/user/blog ;  
USER=root ; COMMAND=/usr/bin/systemctl restart blog  
/var/log/auth.log.4.gz:Aug  3 18:02:51 router sudo: user : TTY=unknown ; PWD=/home/user ; USE  
R=root ; COMMAND=/usr/bin/systemctl stop blog  
/var/log/auth.log.4.gz:Aug  3 18:02:51 router sudo: user : TTY=unknown ; PWD=/home/user/blog  
; USER=root ; COMMAND=/usr/bin/systemctl start blog  
/var/log/auth.log.4.gz:Aug  3 18:03:44 router sudo: user : TTY=pts/0 ; PWD=/home/user ; USER=  
root ; COMMAND=/usr/bin/systemctl stop blog  
/var/log/auth.log.4.gz:Aug  3 18:03:48 router sudo: user : TTY=pts/0 ; PWD=/home/user ; USER=  
root ; COMMAND=/usr/bin/systemctl start blog  
/var/log/auth.log.4.gz:Aug  3 18:04:06 router sudo: user : TTY=pts/0 ; PWD=/home/user ; USER=  
root ; COMMAND=/usr/bin/systemctl stop blog  
/var/log/auth.log.4.gz:Aug  3 18:05:52 router sudo: user : TTY=unknown ; PWD=/home/user ; USE  
R=root ; COMMAND=/usr/bin/systemctl stop blog  
/var/log/auth.log.4.gz:Aug  3 18:05:52 router sudo: user : TTY=unknown ; PWD=/home/user/blog  
; USER=root ; COMMAND=/usr/bin/systemctl start blog  
/var/log/auth.log.4.gz:Aug  3 18:06:19 router sudo: user : TTY=pts/0 ; PWD=/home/user ; USER=  
root ; COMMAND=/usr/bin/systemctl start blog  
/var/log/auth.log.4.gz:Aug  3 19:35:34 router sudo: user : TTY=unknown ; PWD=/home/user ; USE  
R=root ; COMMAND=/usr/bin/systemctl stop blog  
/var/log/auth.log.4.gz:Aug  3 19:35:34 router sudo: user : TTY=unknown ; PWD=/home/user/blog  
; USER=root ; COMMAND=/usr/bin/systemctl start blog  
user@router:~$
```

Part 2: Add remediation tasks

6. Make a screen capture showing your task description.



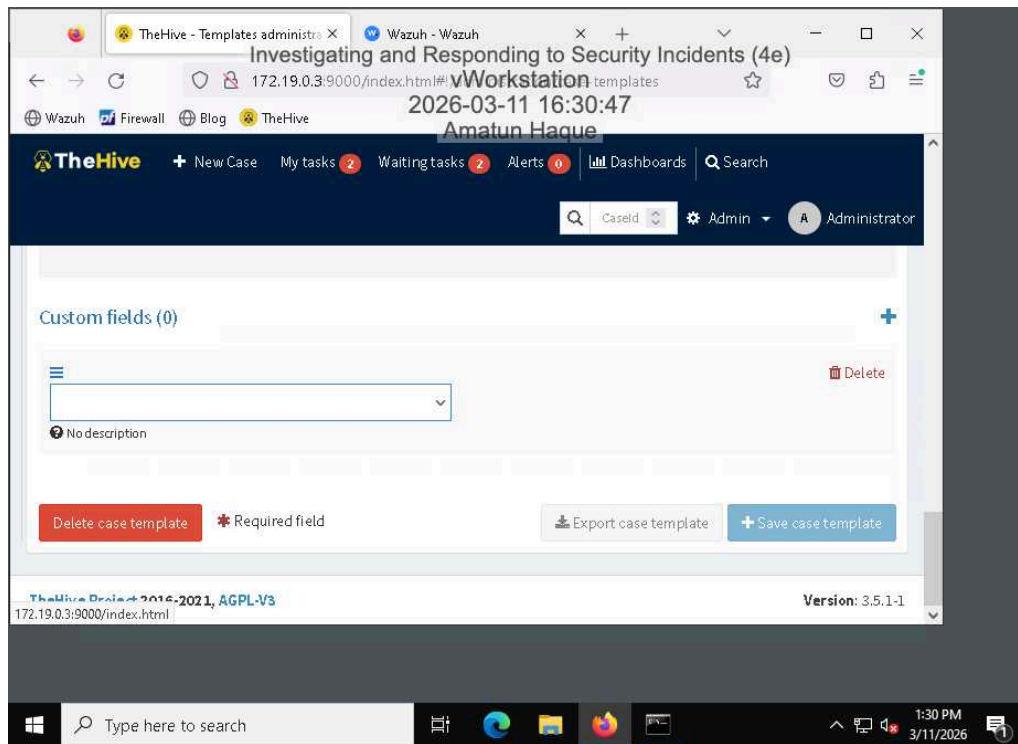
9. Make a screen capture showing your task description.



Challenge and Analysis

Part 1: Add real time monitoring for nginx load balancer configuration

1. Make a screen capture showing the updated group configuration.



Part 2: Disable the compromised key

1. Document all changes made to the SSH configurations and any other steps required.

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Student:	Email:
Amatun Haque	amatunh.student@wust.edu

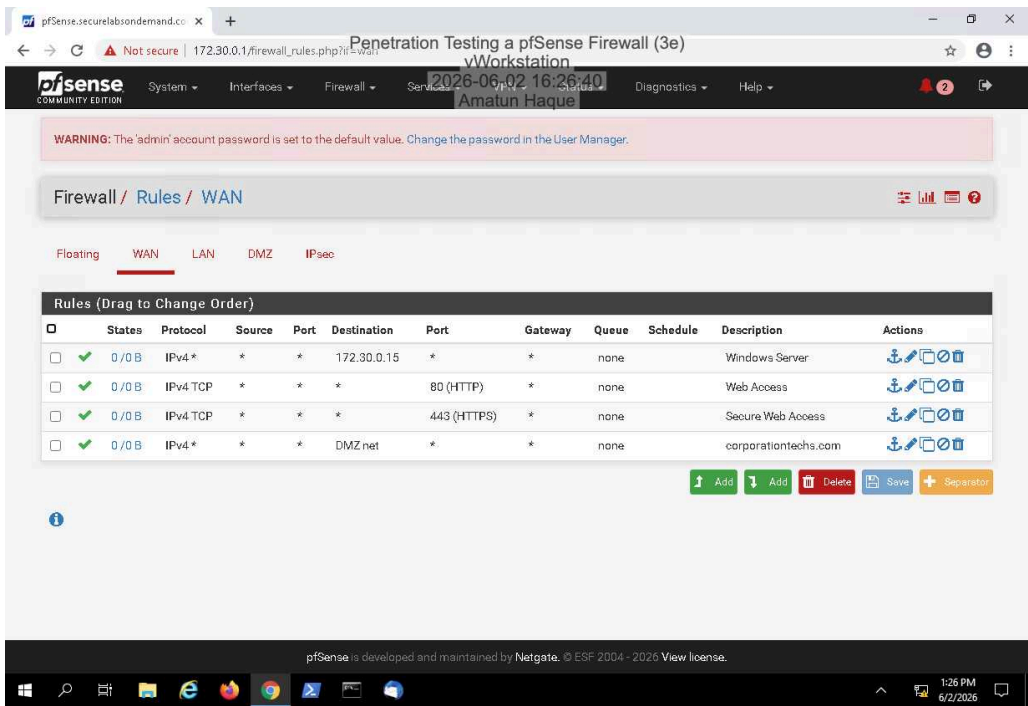
Time on Task:	Progress:
8 hours, 10 minutes	100%

Report Generated: Tuesday, June 2, 2026 at 5:00 PM

Section 1: Hands-On Demonstration

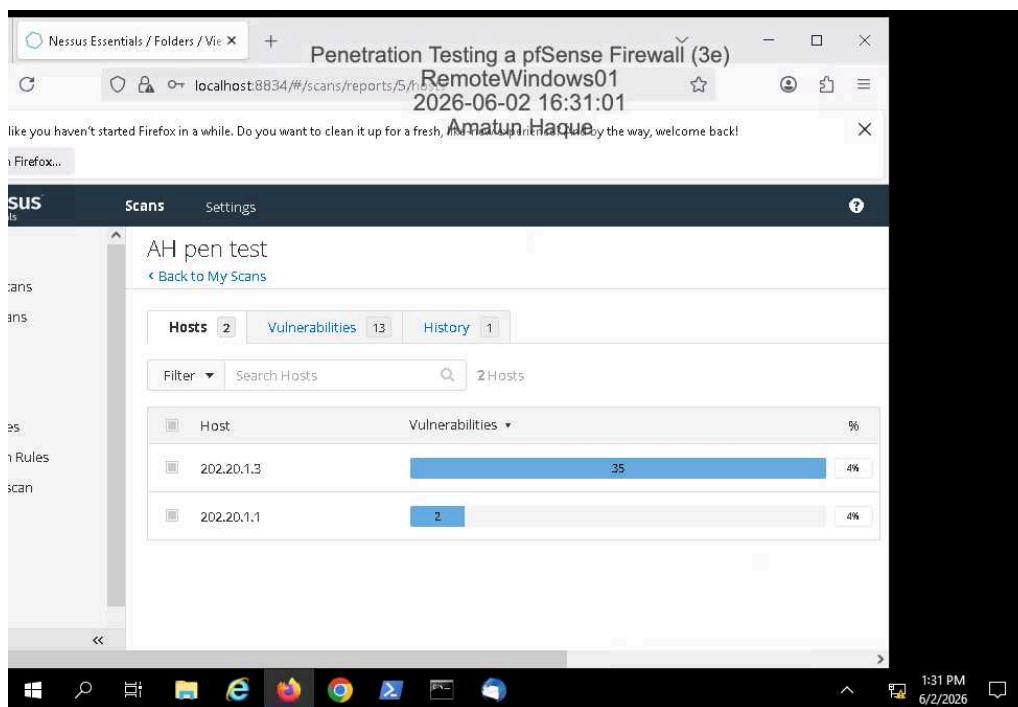
Part 1: Examine a pfSense Firewall Configuration

12. Make a screen capture showing the WAN rules table.

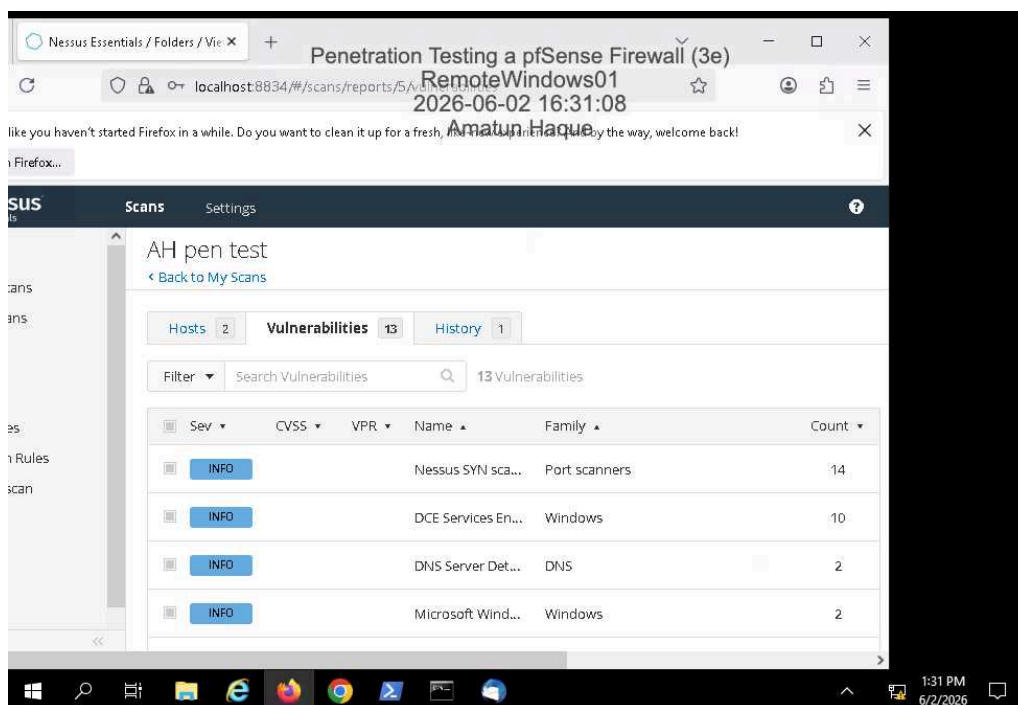


Part 2: Conduct a Penetration Test on the Network

11. Make a screen capture showing the *yourname* pen test scan results.



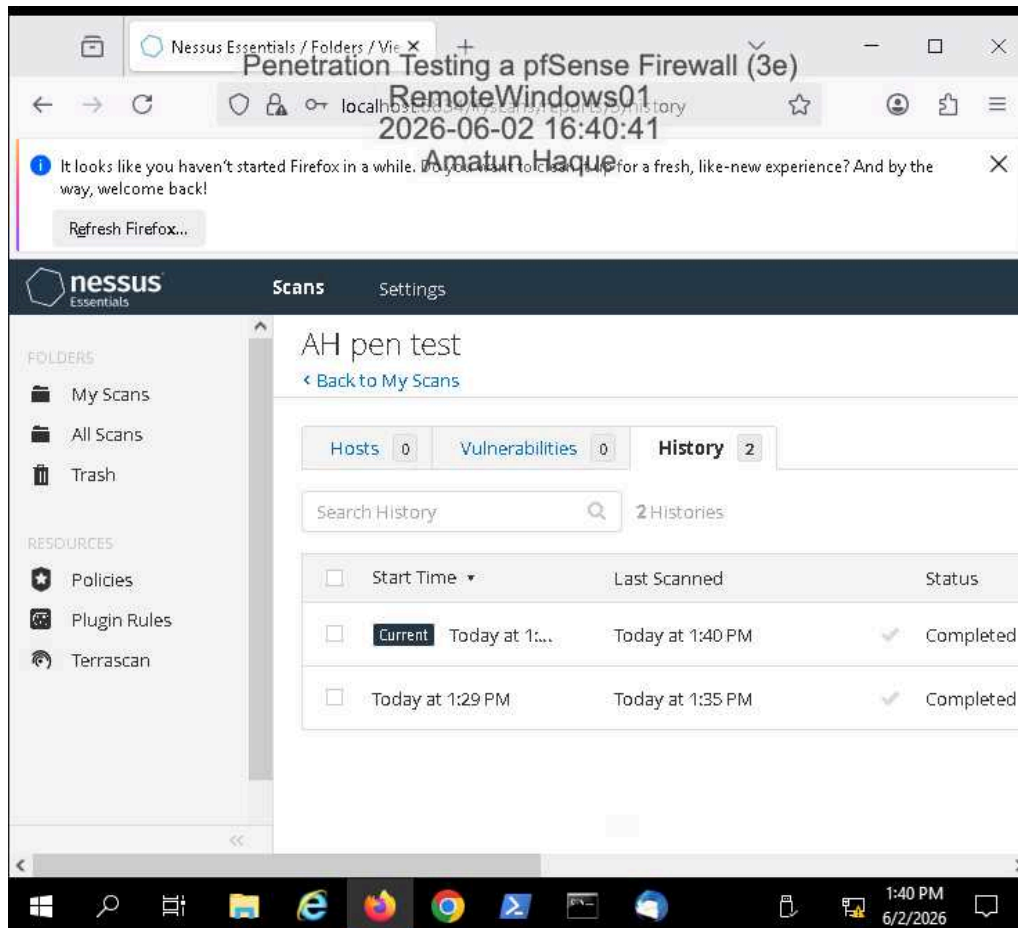
13. Make a screen capture showing the list of vulnerabilities.



Penetration Testing a pfSense Firewall (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 10

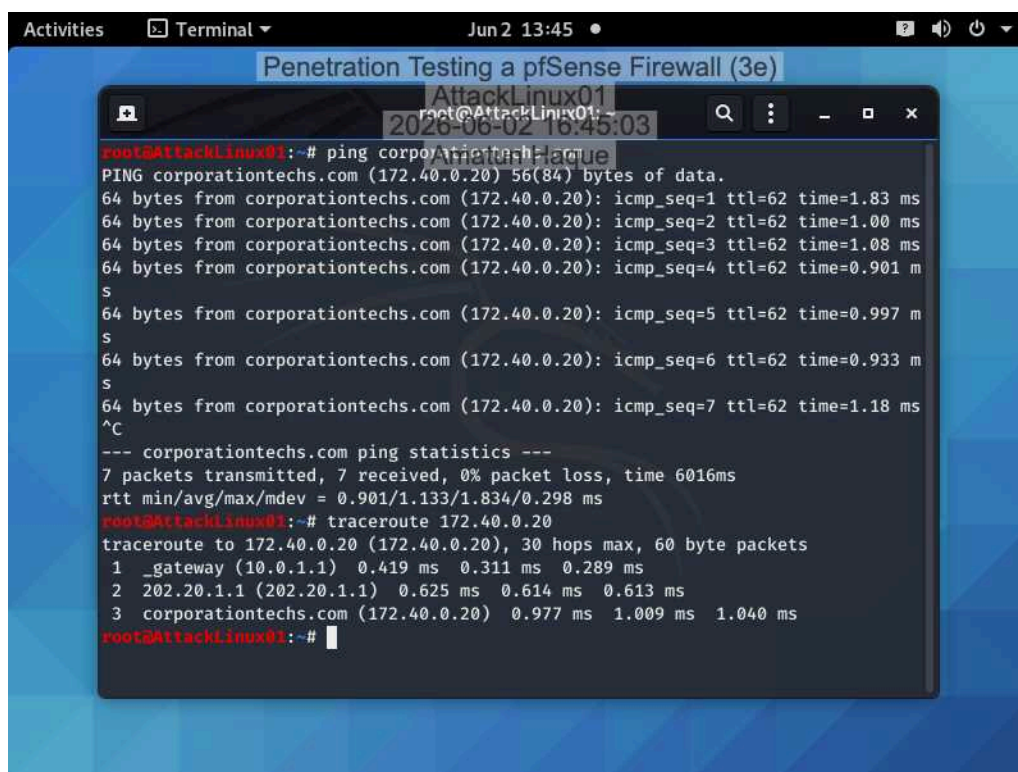
30. Make a screen capture showing the **updated vulnerability report summary**.



Section 2: Applied Learning

Part 1: Conduct a Port Scan on the Network

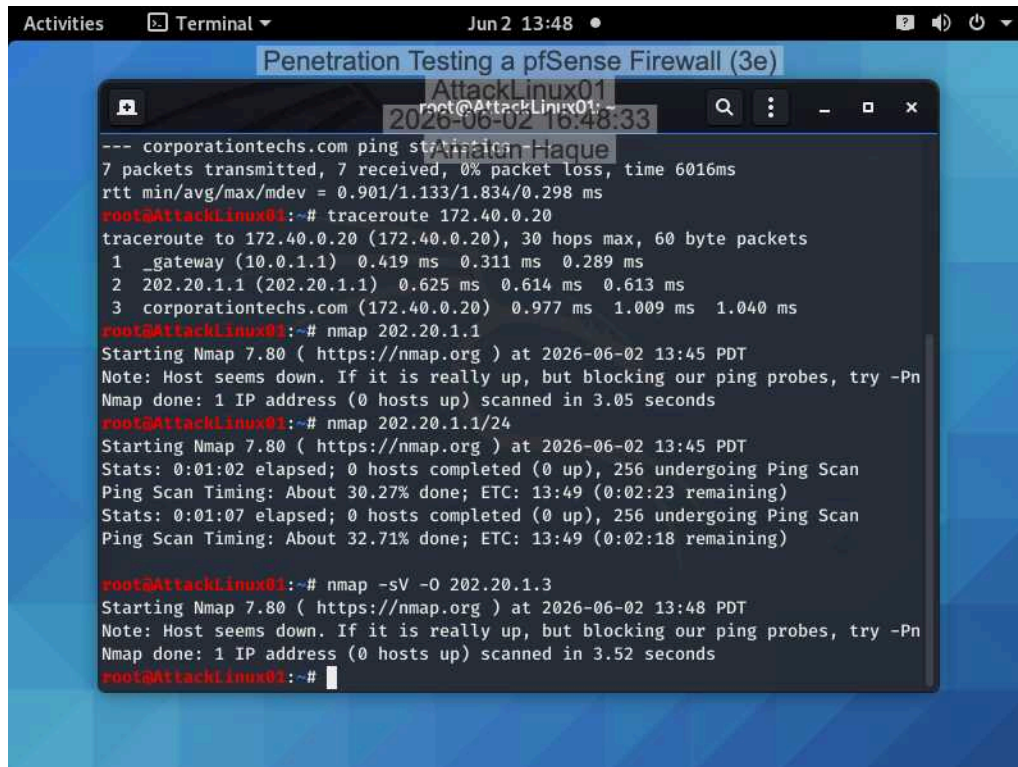
7. Make a screen capture showing the results of the traceroute command.



The screenshot shows a terminal window titled "AttackLinux01" with a date and time stamp of "2026-06-02 16:45:03". The terminal output shows the results of a ping command to corporationtechs.com (172.40.0.20) and a traceroute to the same destination. The ping command shows 7 successful packets with varying round-trip times. The traceroute shows three hops: the first hop is the gateway (10.0.1.1), the second hop is 202.20.1.1, and the third hop is corporationtechs.com (172.40.0.20).

```
root@AttackLinux01:~# ping corporationtechs.com
PING corporationtechs.com (172.40.0.20) 56(84) bytes of data:
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=1 ttl=62 time=1.83 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=2 ttl=62 time=1.00 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=3 ttl=62 time=1.08 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=4 ttl=62 time=0.901 m
s
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=5 ttl=62 time=0.997 m
s
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=6 ttl=62 time=0.933 m
s
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=7 ttl=62 time=1.18 ms
^C
--- corporationtechs.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6016ms
rtt min/avg/max/mdev = 0.901/1.133/1.834/0.298 ms
root@AttackLinux01:~# traceroute 172.40.0.20
traceroute to 172.40.0.20 (172.40.0.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 0.419 ms 0.311 ms 0.289 ms
 2 202.20.1.1 (202.20.1.1) 0.625 ms 0.614 ms 0.613 ms
 3 corporationtechs.com (172.40.0.20) 0.977 ms 1.009 ms 1.040 ms
root@AttackLinux01:~#
```

11. Make a screen capture showing the result of the nmap scan with OS detection activated.



The screenshot shows a terminal window titled "Penetration Testing a pfSense Firewall (3e)" with a window manager overlay. The terminal is running on a system named "AttackLinux01" with IP "172.40.0.20". The user is root. The following commands and their outputs are shown:

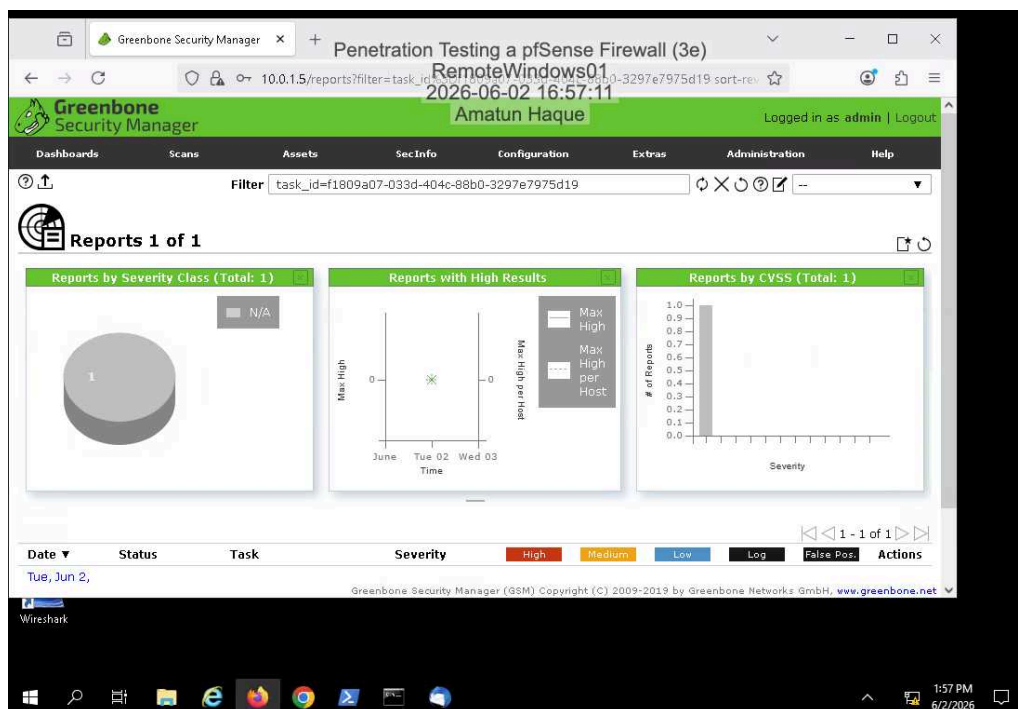
```
--- corporationtechs.com ping statistics
7 packets transmitted, 7 received, 0% packet loss, time 6016ms
rtt min/avg/max/mdev = 0.901/1.133/1.834/0.298 ms
root@AttackLinux01:~# traceroute 172.40.0.20
traceroute to 172.40.0.20 (172.40.0.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1)  0.419 ms  0.311 ms  0.289 ms
 2 202.20.1.1 (202.20.1.1)  0.625 ms  0.614 ms  0.613 ms
 3 corporationtechs.com (172.40.0.20)  0.977 ms  1.009 ms  1.040 ms
root@AttackLinux01:~# nmap 202.20.1.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-02 13:45 PDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.05 seconds
root@AttackLinux01:~# nmap 202.20.1.1/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-02 13:45 PDT
Stats: 0:01:02 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 30.27% done; ETC: 13:49 (0:02:23 remaining)
Stats: 0:01:07 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 32.71% done; ETC: 13:49 (0:02:18 remaining)
root@AttackLinux01:~# nmap -sV -O 202.20.1.3
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-02 13:48 PDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.52 seconds
root@AttackLinux01:~#
```

Part 2: Conduct a Vulnerability Scan on the Network

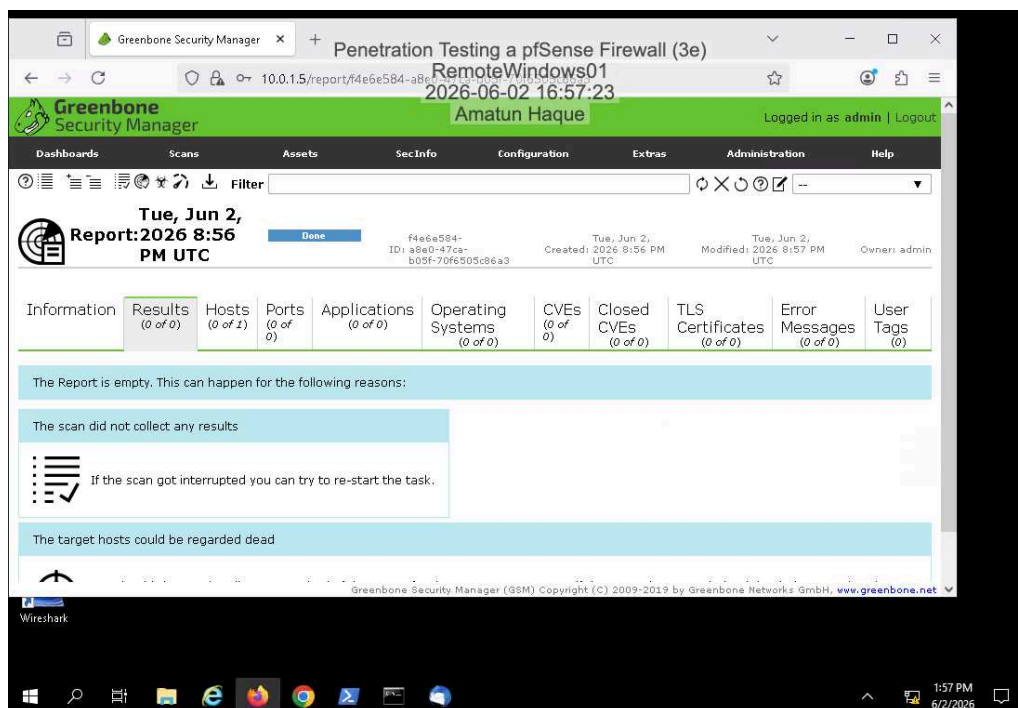
Penetration Testing a pfSense Firewall (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 10

12. Make a screen capture showing the OpenVAS scan report.



14. Make a screen capture showing the detailed OpenVAS scan results.



Section 3: Challenge and Analysis

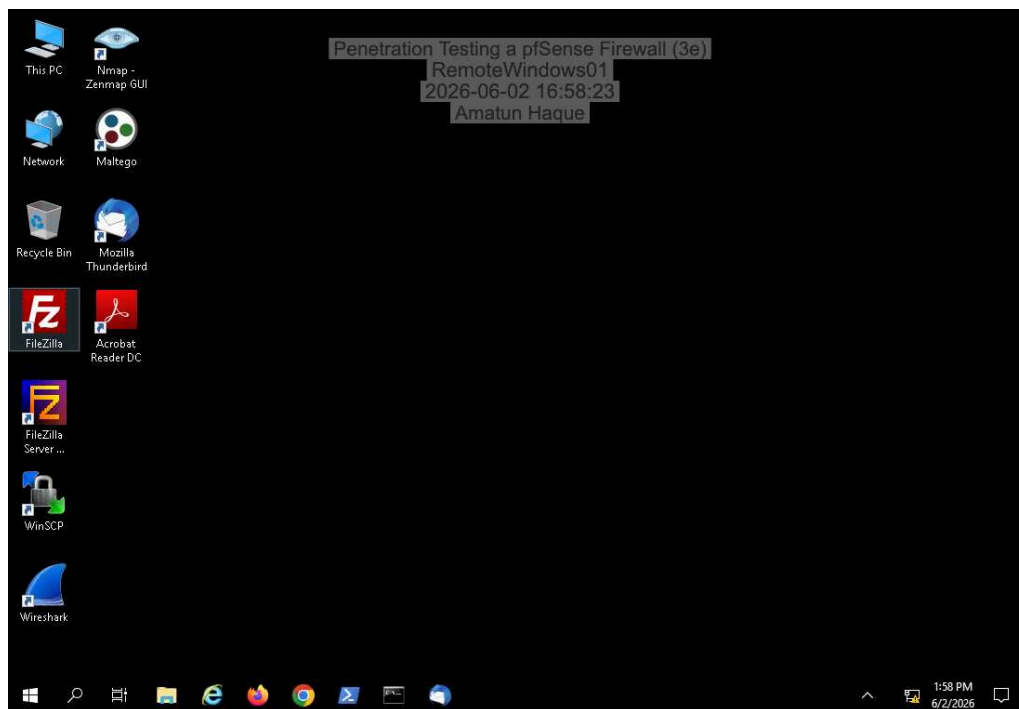
Part 1: Research DMZ Deployment Best Practices

Before beginning the technical portion of your penetration test, you decide to spend some time brushing up on best practices and common mistakes for DMZ deployments - both the network aspect and the servers located therein. Use the Internet to **research** DMZ deployments, then **identify** three best practices and one potential mistake or vulnerability.

?

Part 2: Conduct a Penetration Test on the DMZ

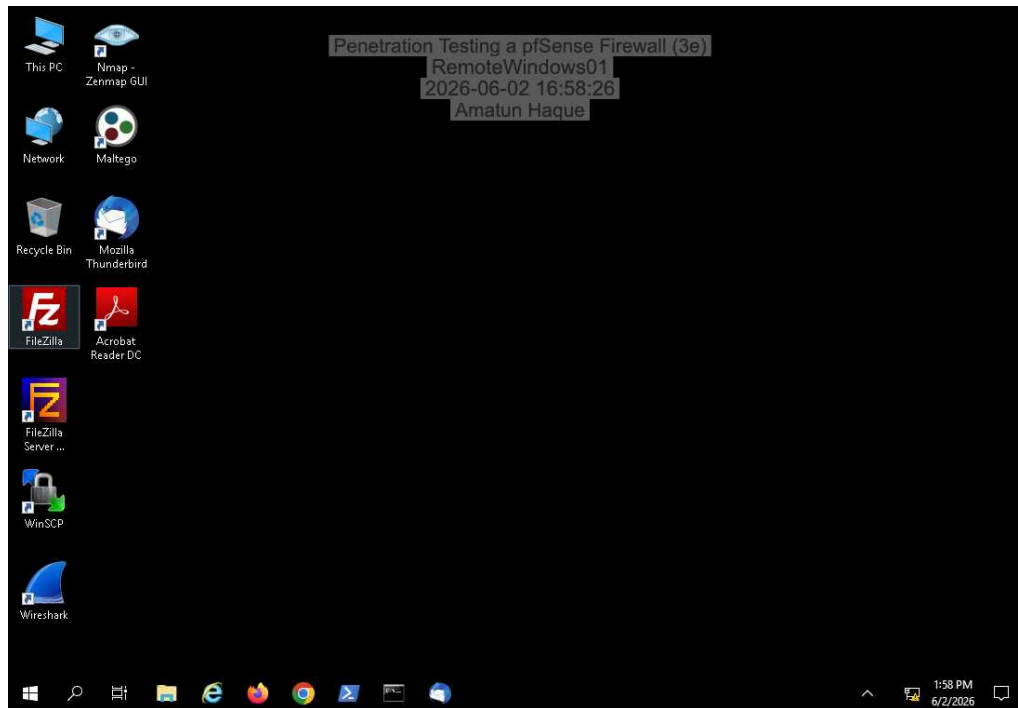
Make a screen capture showing the open ports on the corporationtechs.com web server and the DMZ firewall interface.



Penetration Testing a pfSense Firewall (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 10

Make a screen capture showing the vulnerability scan results.



Part 3: Recommend Changes to the DMZ

Based on your research in Part 1 and your findings in Part 2, **prepare a brief summary** of recommended changes that Secure Labs on Demand should make to their DMZ deployment. Remember, your recommendations should apply to both the network configuration and the web server.

?